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THE EFFECT OF EMPHASIZING PROOFREADING AND SELECTED
BASIC ENGLISH RULES IN A BEGINNING TYPEWRITING
CLASS AT THE COLLEGE LEVEL

by

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CHAPTER I

INTRODUCTION

Among the major contemporary problems in business education listed by Smith in 1977 is one confronting all areas of education: "Those who enroll in business courses exhibit an amazing deficiency in reading, writing, and computing skills."¹ To help alleviate this national problem, an interdisciplinary instructional approach is needed to aid students in achieving more effective communication skills. Teachers are encouraged to explore possibilities in their individual courses for including constructive practice in oral and written expression.

Statement of the Problem

Since typewriting is a communication skill in itself by virtue of providing copy faster and easier to read than longhand, the course affords many challenging opportunities for developing skills related to typewritten messages. Bennett has stated, "Typewriting is the prime course in the business curriculum for the integration of

¹Genevieve Smith, Techniques and Activities for Teaching Business Skills--Featuring the Mini-Company Approach (West Nyack, New York: Parker Publishing Company Inc., 1977), p. 5.

communication skills."¹ According to Erickson, research evidence indicates that the typewriter is a relatively unexploited tool as an aid to learning.² There is considerable evidence that typewriting improves facility in language arts if it is taught for that purpose. Ancillary benefits from learning to typewrite include improved spelling, punctuation, vocabulary, composition, organization, and other language skills. In writing of the interdisciplinary nature of typewriting, Grill points out the contributions that the disciplines of English, psychology, and mathematics have made to the study of typewriting.³ Conversely, he recognizes that many significant contributions can be made to other subjects by typewriting. According to Bennett, "Our challenge is clear, but we cannot expect communication skills to be learned only incidentally. They must be programmed and planned as a definite part of the instructional pattern . . ."⁴

Any statement of overall objectives of the typewriting course normally includes related learnings necessary for effective written communication. For example, the

¹James C. Bennett, "Integrate Communication Skills Into the Secondary Curriculum," Business Education Forum 32 (November 1977):29.

²Lawrence W. Erickson, "Using the Typewriter for Learning: Writing," Balance Sheet (December 1976/January 1977), p. 149.

³George P. Grill, "Interdisciplinary Nature of Typewriting," Business Education Forum 19 (December 1964): 19-20.

⁴Bennett, "Integrate Communication Skills," p. 29.

fifth objective of typewriting enumerated by the Georgia Department of Education is as follows: "To increase students' ability to spell, punctuate, syllabicate, and to compose good English sentences and to integrate these related learnings with typewriting skills."¹ In actual practice, however, Hertz surmised:

In terms of accomplishments by the students, emphasis in the classroom and practice by the teachers, the objectives of typewriting may be broken down into two categories--primary and secondary.

It would seem that the primary objectives--those accomplished the most, emphasized the most, and practiced the most--are concerned with the following areas of typewriting: parts of the typewriter, typewriting techniques, margins, straight-copy speed and accuracy, centering and tabulation, business letters, and manuscripts and outlines.

It also seems that the secondary objectives--those accomplished the least, emphasized the least, and practiced the least--are concerned with the following areas of typewriting: composition, proofreading, erasing, problem solving, attitude, work habits, and mailable copy.²

Regarding composition at the typewriter, in 1968

Cook submitted:

When typewriting was first introduced into the public school curriculum, it served two primary purposes: (1) to provide senior high school students with "touch" copying skills and (2) to provide elementary school students with a greater facility in reading and composition. . . . Today, many teachers extol both virtues in teaching typewriting. Most, however, put the major emphasis on copying rather than writing at the typewriter. A substantial amount of research over

¹Allien R. Russon and S. J. Wanous, Philosophy and Psychology of Teaching Typewriting, 2nd ed. (Cincinnati: South-Western Publishing Company, 1973), pp. 25-26.

²Daniel G. Hertz, "Are We Making Robots Out of Our Typewriting Students?" Business Education Forum 27 (December 1972):26.

the past 45 years clearly demonstrates that a teacher who is so inclined can accomplish three objectives in her typewriting classes:

- Teach the basic principles of touch typing
- Teach the student how to use the typewriter as a writing tool
- Teach content--i.e., spelling, economics, language arts, and so forth¹

In the considered opinion of Tonne, Popham, and Freeman:

. . . the expected outcomes for a typewriting course vary from school to school, from teacher to teacher, and from student to student. There are, however, basic objectives identifiable for all beginning typewriting instruction. Their emphases vary in degree with the various situations in which typewriting is taught, but not in basic essentiality.²

Additional research is needed to determine ways of integrating communication skills in the typewriting course and the effect of emphasizing these skills on the fundamental acquisition of rapid and accurate typewriting skills. While related skills are customarily included to some degree in advanced typewriting courses, the greater contribution to the goal of improving written communication skills among students would be found in beginning classes where substantially larger numbers of students are enrolled.

Over five million students study typewriting each year in middle and elementary schools, junior and senior

¹Fred S. Cook, "Use 'Thought Starters' for Composing at the Typewriter," Business Education World, June 1968, pp. 12-13, 19-20.

²Herbert A. Tonne, Estelle L. Popham, and M. Herbert Freeman, Methods of Teaching Business Subjects (New York: Gregg Division, McGraw-Hill Book Company, 1965), p. 116.

high schools, postsecondary institutions, private business schools, and adult courses.¹ Seven of ten students receive one year or less of typewriting instruction.² The largest typewriting enrollments are found in senior high school grades ten, eleven, and twelve.³ Typewriting enrollment in secondary schools ranks second only to the enrollment in the required English courses.⁴ Typewriting provides more job opportunities for students than any other single course.⁵ Because of increased enrollments of both nonvocational and vocational students, Tonne strongly suggests that teachers of typewriting "examine and reexamine the changes in subject matter and methodology demanded by the situation."⁶

A major component of proficiency in the skill of typewriting is proofreading. However, according to Gryder and Gryder:

. . . proofreading has become one of the neglected areas of instruction in typewriting. . . . A first-year business teacher in a large secondary school reports that the major problem with proofreading is that students do not recognize errors as errors. They do not

¹John L. Rowe and Laverne Thoreson, "A Historical Development of Typewriting," Business Education--Yesterday, Today, and Tomorrow, National Business Education Association Yearbook, No. 14, 1976, p. 106.

²Ibid., p. 107.

³Ibid., p. 106.

⁴James R. Meehan, "Trends in the Teaching of Typewriting," Journal of Business Education 52 (April 1977): 308.

⁵Ibid., p. 307.

⁶Tonne, Popham, and Freeman, Methods, p. 112.

know how to spell properly, to punctuate, to apply rules of grammar, and indeed cannot distinguish many of these errors. . . . A variation on the theme expressed by collegiate typewriting teachers is that students do not take time to proofread carefully. . . . A major responsibility of the typewriting instructor on the collegiate level--or indeed any level--is to make certain that students develop the habit of proofreading everything they type.¹

Wong reported that a survey undertaken by Ewart in the 1960s provided supporting evidence for widespread proofreading disability.² A 369-word test containing thirty-eight errors was given to 2,538 subjects. Only one person found all the mistakes. Table 1 ranks the subjects by groups according to their mean total error scores:

TABLE 1
PROOFREADING SURVEY SUMMARY

Group	Mean Total Errors
Business Teachers	8.9
Professional Secretaries	9.8
College Students	12.0
Miscellaneous Adults	13.2
High School Students	13.3
Junior High School Students	15.9
Mean % inaccuracy level for all groups:	32%

Source: Margaret J. Ewart, Unpublished study to determine types of straight copy proofreading errors made by typewriting students and others (Corvallis, Oregon State University, 1963).

¹Robert Gryder and Isabelle Gryder, "Proofreading: A Neglected Area of Typewriting Instruction," Business Education Forum 30 (October 1975):18-19.

²Shirley Wong, "What Are We Doing About Proofreading," Journal of Business Education 49 (December 1973):122.

According to Wise, proofreading includes locating and correcting typographical errors as well as errors in (1) content, (2) format, (3) grammar, (4) spelling, (5) capitalization, (6) punctuation, (7) word usage, and (8) word division.¹ Proofreading, then, brings into focus four basic English components which are particularly appropriate to stress in the typewriting course: (1) capitalization, (2) punctuation, (3) spelling, and (4) word division. The problem of instructional planning and methods is to stress these components without sacrificing maximum development of rapid and accurate keyboard manipulation. Crawford has pointed out:

The purpose of typewriting instruction should not be diluted or minimized. Regardless of the title of the course being offered, the primary goal of any course in typewriting is mastery of the machine called "typewriter." No amount of allusion to other purposes--to incidental or concomitant learnings--can alter the fact that the chief aim of typewriting courses is to teach students how to typewrite most efficiently.²

Further research is needed to aid in the development of systematic methods of emphasizing proofreading and selected basic English rules in beginning typewriting and to determine the effect of this emphasis. This need to aid students in achieving more effective writing skills is the problem which prompted the study.

¹Elva Lea Wise, "Proofreading is for Everyone," Balance Sheet, December 1974/January 1975, p. 172.

²T. James Crawford, "Fundamental Considerations for Successful Typewriting Instruction," Business Education Forum, 12 (October 1962):20.

Need for the Study

Declining national test scores

In October, 1975, the National Assessment of Educational Progress (NAEP) Report on Writing Mechanics for 1969-1974¹ was released, showing a decline in literacy skills which correlated with declining average national test scores in the Scholastic Aptitude Test (SAT) and American College Test (ACT) since 1964. The development of NAEP was, according to Shafer, a response to the public's growing concern about the quality of education.² The Seventh Annual Gallup Poll of Public Attitudes Toward Education, conducted in June, 1975, revealed that the third major reason in the opinion of the respondents for declining national test scores was "not enough emphasis on basics," following (1) "students' lack of interest/motivation" and (2) "lack of discipline in the home and school."³

In 1976, in the Eighth Annual Gallup Poll, 59 percent of the respondents believed that the decline in national test scores in recent years meant that the quality

¹National Assessment of Educational Progress, Writing Mechanics 1969-1974: A Capsule Description of Changes in Writing Mechanics (Washington: U. S. Government Printing Office, October 1975).

²Robert E. Shafer, "National Assessment: Backgrounds and Projections--1975," Paper presented at the Annual Meeting of the Conference on English Education, Colorado Springs, Colorado, 1975 (ERIC Document Reproduction Service, ED 108 211, 1975).

³George H. Gallup, "Seventh Annual Gallup Poll of Public Attitudes Toward Education," Phi Delta Kappan 57 (December 1975):236.

of education was declining.¹ The 1976 poll also showed that the respondents' top choice of ways to improve the quality of public education was "devoting more attention to the teaching of basic skills."²

After the NAEP report was issued, Newsweek (December 8, 1975) published the cover story, "Why Johnny Can't Write," which was read by over two million Americans. The article exposed the current condition of the teaching of writing. It purported that the present state of English skills would not qualify high school graduates to undertake college work nor enable clerical workers to hold jobs. Moreover, the article pointed out that some college students were graduating although semi-illiterate. This story fit the national mood of the 1970s, and its ready acceptance paralleled that of its predecessor, Rudolf Flesch's Why Johnny Can't Read³ of the 1950s.

Literacy "crisis"

A government report released in October, 1975, pertaining to a four-year study at the University of Texas documented the fact that one of every five American adults is functionally illiterate and unable to cope effectively with daily duties and activities. The report also disclosed

¹George H. Gallup, "Eighth Annual Gallup Poll of the Public's Attitude Toward the Public Schools," Phi Delta Kappan 58 (October 1976):190.

²Ibid., p. 189.

³Rudolf Franz Flesch, Why Johnny Can't Read--And What You Can Do About It (New York: Harper, 1955).

that one out of three adults functions on a minimal level in the nine areas of the knowledges and skills tested: occupational information, consumer economics, government and law, health, community resources, reading, problem-solving, computation, and writing.¹

Other examples of a "literacy crisis" in public education have been noted:

1. In recent years more than one-third of the applicants to the journalism school of the University of Wisconsin at Madison have not been able to meet the minimum requirements in spelling, grammar, word usage, and punctuation²
2. In the University of California system, half of those admitted as freshmen in 1974 were required to take a remedial English course. The freshman class consisted of the top eight percent of California high school graduates³
3. Thirty to fifty percent of applicants to journalism programs at Minnesota, Texas, and North Carolina have recently failed basic spelling, punctuation, and usage tests⁴
4. Twelve to fifteen percent of all American students fail to read beyond an elementary level⁵

¹William W. Brickman, "Adult Functional Competency and Educational Malfunctioning," Intellect 104 (March 1976): 451.

²Vermont Royster, Wall Street Journal, cited by George Gallup, "Public Schools: Their Problems, Their Future," NASSP Bulletin (May 1975):5.

³Glen A. Lyons, "The Higher Illiteracy," Harper's, September 1976, p. 34.

⁴Ibid.

⁵Ibid.

5. Only half the seventeen-year-olds in the United States can read college freshman-level materials.¹
6. Researchers Lipset and Ladd surveyed 4,400 professors at 161 colleges and reported that virtually all faculty members said that many students with whom they come in close contact "are seriously underprepared in basic skills, such as those required for written and oral communication"²
7. On the Minnesota Scholastic Aptitude Test, which is given to 90 percent of high-school juniors in Minnesota, scores have declined steadily since the mid-1960s; Iowa Tests of Educational Development have shown similar results³

Dissatisfaction in business sector

In addition to outspoken members of the press and college faculties, the business sector has been equally vocal in expressing dissatisfaction with the lack of communication skills displayed by the current generation. Erickson has testified:

Those who employ graduates of high schools decry the inability of workers to prepare memorandum reports, letters, and formal reports that are understandable and free of glaring errors in syntax, rhetoric [sic], and mechanics. Among the most frequently mentioned weaknesses in writing ability are: inaccurate use of words, inability to spell correctly, poor punctuation skill, inability to apply correctly some of the most basic grammar rules, weak capitalization skill, lack of

¹Malcolm G. Scully, "Only Half of 17-Year-Olds Found Able to Read Well," Chronicle of Higher Education, 14 (28 March 1977):2.

²"Paychecks, Dropping Enrollments Causing Campus Professor Blues," Washington Post, reprinted in Tuscaloosa News, 9 January 1978, p. 8.

³Malcolm G. Scully, "What's Causing the Drop in College Entrance Scores?" Chronicle of Higher Education 11 (17 February 1976):3.

consistency in expressing numbers, and inability to compose simple messages.¹

In 1975 the Advisory Council for Technical-Vocational Education in Texas completed a statewide survey, "Qualities Employers Like, Dislike, in Job Applicants."² Employers were asked to respond concerning entry-level jobs requiring high school education. Two of ten leading reasons employers gave for rejecting applicants after the job interview were: (1) "inability of applicant to communicate during the interview" and (2) "poorly filled out job application form." When asked to cite areas of improvement needed in education to prepare for entry into full-time work, employers ranked in seventh place "the ability to write and speak effectively."

In 1976 the National Secretaries Association administered an examination to 746 business education students at high school and post-secondary levels throughout the United States for the purpose of selecting the recipient of the "Miss Future Secretary Award." The scoring of the examination implied that nationwide business students need more work on vocabulary, particularly in the area of word relationships, meaning of words, and word division.³

¹Erickson, "Using the Typewriter for Learning: Writing," p. 148.

²Kenneth W. Brown, "What Employers Look For in Job Applicants," Business Education Forum 30 (April 1976):7.

³Jerre E. Gratz, "Analysis of Examination for Future Secretaries," Journal of Business Education 51 (May 1976): 370.

In a survey of twenty-five executives chosen by random sample from "Fortune 500" corporations, Hammer found that the majority of those interviewed cited a need for improved basic skills and the development of accuracy.¹ Hammer stated that managers "expect careful proofreading."

Functional literacy

Beginning in 1972 nine major committees, councils, panels, task forces, and agencies were commissioned to conduct inquiries into the condition of the American public high school. Among those were the Carnegie Commission on Higher Education, the National Commission on the Reform of Secondary Education, and The Task Force on Secondary Schools in a Changing Society. At no other time in history have so many statements on needed improvements in secondary education been issued in a brief period of less than four years.² Bruce, Firth, Grimsley, and Halverson reviewed the proposals and recommendations of the nine committees and organized them into categories. Concerning program, the reviewers found the recommendation to "establish improvement of basic skills" as a top priority. The recommendation was expanded to suggest the provision of resources, incentives,

¹George Hammer, "The Outlook for Office Jobs: A Survey of the Fortune 500," Delta Pi Epsilon Journal, 18 (November 1975):1-5, as cited by Fred S. Cook and Frank W. Lanham, "R & D," Business Education World, (May-June 1976), p. 15.

²Roy E. Bruce, et al., "A Review of Recent Proposals for Reform in Secondary Education," Educational Forum 40 (March 1976):276.

and rewards for teachers who commit themselves to this task. Of the report from the National Commission on the Reform of Secondary Education, the review stated:

There is a clear urgency implied in these goal statements: The high school has the obligation to certify that every graduate has mastered the skills of reading and writing to a level of functional literacy . . . and the skills of computation to a level sufficient for the management of household responsibilities.¹

College writing skills

Concerning writing skills at the college level, U. S. Education Commissioner Ernest L. Boyer deplored in an Associated Press release:

I cannot believe sometimes the sloppy sentences that are found in correspondence or in college compositions. . . . Direct, clear, forceful expression is a reflection of clear and uncluttered thinking. I'm convinced that American education must once again focus on this essential goal.²

William D. Schaefer, executive secretary of the Modern Language Association, concurred with Boyer, "My impression is that on a national level we have failed--have continued to fail--to meet the challenge of illiteracy among college-level students."³

Dr. L. Edward Mullins, of The University of Alabama School of Communication, has pointed out:

Symptoms of poor writing are visible in all areas of study. . . . The problems won't have the improvement we

¹Ibid., p. 149.

²"Educator Assails Jargon," Tuscaloosa News, 2 December 1977, p. 12.

³"Crisis in English Writing," Chronicle of Higher Education 9 (September 23, 1974):1, 6.

want until there is more emphasis at the elementary, junior high, and high school levels on writing, spelling, and grammar. Right now, students are coming to college so handicapped in these areas they are having to back up and learn to write before they can go on with their studies.¹

Robert Farrell, dean of writing at Cornell University stated, "The writing decline is not only marked, but it's massive. This is not a national problem; it's an international one. My colleagues in Sweden and Germany are experiencing the same thing we are here."²

Scully reported that after the media stepped up its focus on the problem, a vast number of programs were instigated to help correct writing deficiencies among high school and college students. However, many English teachers agree with Astro of Oregon State University who contends that "the responsibility for teaching writing should lie with the entire faculty, not just professors of English."³

English across the curriculum

Although it is generally conceded that the "writing crisis" is no new development, current demands for accountability, relevance, and vocational competencies make the time right for attention to the problem and concentrated effort for remedial action. The current situation places

¹"Poor Writing Said National Problem," Tuscaloosa News, 31 October 1977, p. 5.

²"Colleges Trying Again to Teach Students How to Write," Tuscaloosa News, 28 September 1977, p. 41.

³"'Writing Crisis' Spurs Big Corrective Effort," Chronicle of Higher Education 13 (18 October 1976):12.

reemphasis on the adage: "Every teacher is a teacher of English." In recent years it has become apparent that there should be more research into the needs of English across the curriculum. Charles Stuart-Jervis has said, ". . . English teaching is a shared responsibility in a way that no other subject is, for it cuts across all boundaries."¹

The need for reinforcement of basic English rules among students suggested a need for this study involving the integration of special emphases in the typewriting course. If it can be determined that part of typewriting class periods can be devoted beneficially to the learning of communication skills, such evidence could be important to the development of programs and materials for improving these skills.

Purpose of the Study

The purpose of the study was twofold: (1) to develop materials to modify the traditional beginning typewriting course at the college level to include special emphasis on proofreading, capitalization, punctuation, spelling, and word division without deleting or deemphasizing any aspect of the traditional course content and (2) to determine the effect of this special emphasis on the achievement of both typewriting skills and proofreading skills in recognizing errors in capitalization, punctuation, spelling, and word division.

¹Charles Stuart-Jervis, "The Most Important Subject," Times Educational Supplement (20 September 1974), p. 59.

The objective of the experimental method of instruction was to develop proficiency in proofreading while learning to typewrite at the beginning level without sacrificing maximum typewriting skill development.

The effect of the special emphasis on proofreading and selected basic English rules was determined by comparing two classes at The University of Alabama. The control class was taught in the traditional manner; the experimental class received supplementary instructional materials integrated with the traditional materials.

Three measurements were obtained to determine the effect of the experimental treatment: (1) mean change scores from pretest to posttest on an instrument measuring proofreading ability to recognize errors in capitalization, punctuation, spelling, and word division; (2) mean scores on a final examination measuring typewriting production skills; and (3) mean scores on timed writings.

Hypotheses

A study hypothesis was formulated to determine the effect of emphasizing proofreading and selected basic English rules in a beginning typewriting class at the college level. It was assumed that if the experimental treatment were effective, scores of the experimental class on an instrument measuring proofreading ability to recognize errors in capitalization, punctuation, spelling, and word division would increase from pretest to posttest. The appropriate measure of central tendency, the mean, would represent the

experimental group score; and the mean change score from pretest to posttest would be compared with that of the control group.

Thus, the basic hypothesis of the study was that the experimental treatment would result in a statistically significant higher mean change score from pretest to posttest on an instrument measuring proofreading ability to recognize errors in capitalization, punctuation, spelling, and word division than the mean change score achieved in the control class.

Secondary hypotheses were formulated to determine the effect of the experimental treatment on the development of typewriting skills. It was assumed that if the experimental group performed as well as did the control group on typewriting tests of production and straight-copy skills, there would be no evidence to support the theory that the experimental treatment adversely affected development of typewriting skills in the experimental group. The second hypothesis was that there would be no significant difference between the mean scores of the experimental and control groups on a final examination measuring typewriting production skills. A third hypothesis was that there would be no significant difference between the mean scores of the two groups on timed writings.

Procedures of the Study

The following procedures were used in conducting the experiment:

1. Prior to the beginning of the semester, a testing instrument measuring proofreading ability to recognize errors in capitalization, punctuation, spelling, and word division was developed to be administered before and after experimental treatment
2. Course outlines for the two classes were prepared
3. A review of literature pertaining to methods of integrating associated learnings in the typewriting course was conducted
4. Supplementary materials were developed
5. Four unit tests for each group and a comprehensive final examination for both groups were planned
6. The pretest was administered to both groups on the second day of class. Late registrants were given the test at a later time
7. The experimental class received special instructional materials, integrated with the traditional course content, emphasizing proofreading, capitalization, punctuation, spelling, and word division
8. The control class received traditional instruction including proofreading and word division to a comparatively limited degree
9. As the course progressed, each class received four different unit tests. Tests in the experimental class included special emphasis on proofreading and selected basic English rules as an increment of the course grade
10. During the last month of the semester, each student submitted two timed writings for grades. In each class the better of the two grades was used to compute a group mean score
11. During the forty-second class meeting, the posttest was administered to both classes. Students who were absent took the test on the day of the final examination. The mean change scores from pretest to posttest were computed for both groups
12. Both classes received the same final examination measuring typewriting production skills. The mean scores were computed for both groups

13. Typographical errors were counted on the final examination papers of both classes for comparison
14. Statistical analysis was applied to the data derived from the pretest-posttest, timed writing, and final examination
15. Findings were analyzed, conclusions drawn, and recommendations offered

Limitations of the Study

Before generalizations can be made from this study, several limitations must be noted.

Since two intact groups were studied in the experiment, there was no control over the method of selection; and treatments could not be randomly assigned. Therefore, findings and conclusions of the study are limited to beginning college typewriting classes composed of students similar in background and experience to those students in the study.

Since the population could not be specified in the study, variables such as motivation, interest, experience, and intelligence could have operated differently for each group.

Variables such as time of administration of tests, tardiness, absences from class, personal problems, and failure to complete classroom and homework assignments could have affected the statistical data.

Certain variables are confounded with teaching method and should be considered as possible reasons for differences in group achievement scores; namely: The two

classes were taught by different teachers at different times. Since the investigator taught the experimental class, part of the reason for achievement gains in the experimental class could be attributed to the "experimenter effect."

Since the same testing instrument was used before and after experimental treatment, students' practice in taking the first test could have favorably affected the second test scores.

Methods of providing emphasis on proofreading and basic English rules of capitalization, punctuation, spelling, and word division were limited to those developed by the investigator.

Measurements used to compare the two groups on knowledge of proofreading and the basic English rules stressed in the experimental class were limited to those measurements provided through the use of the testing instrument developed by the investigator.

Measurements of performance of production skills used in comparing the two groups were limited to those provided through the use of the particular examination employed.

Assumptions

The following assumptions were basic to the planning and development of the study:

1. Intentional learning methods are more reliable than incidental learning methods

2. The grading structure of the typewriting course can influence the intentional learning of integrated material

Definition of Terms

The following definitions are presented to facilitate reading and understanding the study:

Ancillary learning - learning which is subordinate or supplementary to primary learning objectives.

Concomitant learning - learning which accompanies the primary learning objectives in a subordinate way.

Corollary learning - learning which naturally accompanies the primary learning objectives.

English mechanics - functional details of language usage.

Experimenter effect - an external threat to validity in experimentation wherein the phenomenon occurs that when the experimenter believes something is going to happen, it often does happen.

Five percent level of significance - the level where an observed difference between means should occur by chance only five times out of one hundred.¹

Incidental learning - learning which occurs unconsciously or by way of interjection or digression.

¹Billy L. Turney and George P. Robb, Research in Education: An Introduction (Hinsdale, Illinois: The Dryden Press, Inc., 1971), p. 93.

Intentional learning - learning by design where all parties concerned understand that learning of the material is expected to take place, and rewards or sanctions are imposed for such learning or lack of learning.

Language skills - abilities to use standard English and to recognize the correct or incorrect use of standard English.

Mailable copy - a finished, usable typewritten product which conforms to standards of arrangement, neatness, and accuracy.

Null hypothesis - "an assertion that there really is no difference between the means of the populations in question."¹

One percent level of significance - the level where an observed difference between means should occur by chance only one time out of one hundred.²

Personal typewriting - nonvocational typewriting instruction which emphasizes composition at the typewriter and the typewriting with acceptable speed and accuracy of essays, outlines, speeches, articles, personal business letters, and other personal papers.³

¹Ibid.

²Ibid.

³S. J. Wanous, "A College Course in Personal Typewriting," Business Education Forum 10 (November 1955): 13-14.

Production - practical application of typewriting ability in producing a quantity of work that meets a pre-determined standard for quality (letters, forms, memorandums, manuscripts, tables, etc.).¹

Reinforcement - assistance to strengthen concepts by supplying additional materials on the subject.

Rejection of the null hypothesis - the statistical decision made that the null form of a study hypothesis is unacceptable because the observed difference between the means is so large that the difference very likely did not happen by chance, as opposed to failure to reject the null hypothesis wherein the observed difference between means is so small that the difference may well have occurred as a result of sampling errors.²

Rhetoric - the art of writing well in prose.

Standard English - that usage of the English language which is practiced in business, news media, and the social mainstream as opposed to dialect used by various subcultural groups.

Statistically significant - a description made after an evaluation that an observed difference was caused by something other than mere chance.³

¹Wilson Ashby, "A Statistical Analysis of Selected Factors in Predicting Achievement in Typewriting," Business Education Forum 9 (January 1955):28-29.

²Turney and Robb, Research in Education: An Introduction, p. 93.

³Ibid.

Statistical significance - a decision made by comparing ratios of mean scores.¹

Straight-copy typewriting - copying without regard to layout except for reasonably regular right margins and without correction of errors.

Syllabication - the process of dividing words into syllables (in typewriting: rules for dividing words at the ends of lines to keep the right margin straight).

Syntax - rules for constructing sentences.

Timed writing - a typewriting test consisting of paragraph material to be typed for a predetermined length of time to yield a measurement, expressed in typewriting words per minute, of the ability to type straight copy.

Touch typewriting - the process of copying or composing by striking the typewriter keys and using the mechanisms properly without looking at the hands or the keyboard.

t-ratio - the proportion of the difference between means to the standard error of a difference between means.²

Twenty percent level of significance - the level where an observed difference between means should occur by chance twenty times out of one hundred.

¹Ibid.

²Ibid.

Vocational typewriting - typewriting instruction with the objective of preparing learners for employment as qualified typists.

Organization

Chapter I consists of a general overview of the problem with which the study deals and includes a statement of the problem, need for the study, purpose of the study, hypotheses, procedures of the study, limitations of the study, assumptions, and definition of terms.

Chapter II consists of a review of selected literature related to the study.

Chapter III contains the methods of procedure used in the experiment.

Chapter IV consists of the analysis of the findings of the study.

Chapter V contains the summary, conclusions, and recommendations.

CHAPTER II

REVIEW OF RELATED LITERATURE

Introduction

The purpose of this chapter is to review selected literature relative to the use of the typewriter as an instructional tool for learnings other than the skill of typewriting. Because of noted deficiencies in students' communication skills, recent professional literature has urged serious consideration of the proposal to use the typewriter as a learning instrument. In the 1976-77 school year, six serial articles appeared in the Balance Sheet showing ways in which instruction in typewriting can improve language skills:

1. "Typewriting as a Communication Skill"¹
2. "Using the Typewriter for Learning: Reading"²
3. "Using the Typewriter for Learning: Writing"³

¹S. J. Wanous, "Typewriting as a Communication Skill," Balance Sheet, October 1976, pp. 57-59, 91.

²Berle Haggblade, "Using the Typewriter for Learning: Reading," Balance Sheet, November 1976, pp. 106-10, 137.

³L. W. Erickson, "Using the Typewriter for Learning: Writing," Balance Sheet, December 1976/January 1977, pp. 148-52, 182.

4. "Using the Typewriter for Learning: Spelling"¹
5. "Using the Typewriter for Learning: Composing"²
6. "Using the Typewriter for Learning: Concepts"³

In general, since the beginning of its inclusion as a school subject, typewriting has been used in three ways to aid the learning of associated knowledges and skills:

(1) The typewriter has been put into the language arts classroom as a means of facilitating improvement in self-expression; (2) special, purposive practice drills and typewriting materials containing content to be learned incidentally while typing have been utilized in the typewriting course; and (3) a combination of special materials and regular classroom instructional methods including graded tests have been employed intentionally in teaching related knowledge within the typewriting course.

The review of the related literature includes selected theses and dissertation studies from 1931 to the present time pertaining to ancillary benefits from learning to typewrite and articles appearing in professional journals and magazines pertinent to the use of the typewriter in acquiring associated knowledges. Articles were found

¹Lloyd Bartholome, "Using the Typewriter for Learning: Spelling," Balance Sheet, February 1977, pp. 196-200.

²James C. Bennett, "Using the Typewriter for Learning: Composing," Balance Sheet, March 1977, pp. 244-49.

³Dean Clayton, "Using the Typewriter for Learning: Concepts," Balance Sheet, March 1977, pp. 258-60, 279.

primarily in two magazines distributed by publishers as a service to business educators and three business education journals.

The review has been divided into the following categories: (1) education of young children, (2) vocational and personal typewriting, (3) spelling, (4) incidental and intentional learning, (5) English as a foreign language, (6) special education, and (7) proofreading.

Typewriting and Young Children

The idea that the typewriter is of value in the education of young children is as old as the machine itself. The first typewriter catalog ever published, which came out in 1874, emphasized the importance of the typewriter in developing pupils' language skills; an advertisement in The Nation (16 December 1875) proclaimed that "no device is comparable to the typewriter for teaching children to spell and punctuate."¹

There was no suitable typewriter available to children until the latter part of 1920, when the first strictly portable typewriter with a standard, four-row keyboard was produced. Previously all the small typewriters had three rows requiring too much manual skill to be well

¹Cecilia E. Unzicker, An Experimental Study of the Use of the Typewriter on Beginning Reading (New York: Bureau of Publications Teachers College, Columbia University, 1934), p. 1.

adapted to use by the average child.¹ As long as typewriters remained bulky and costly, scientific experimentation to verify the advantages of using the typewriter as an educational instrument was not attempted. By 1925 the four leading typewriter manufacturers were producing portables; and experimentation was begun shortly thereafter showing the typewriter to have values when used by primary and intermediate grades, such as providing an early and rapid means of writing, a means of producing orderly papers, and an introduction to a useful skill.²

In 1931 Wood and Freeman completed a two-year study with nearly 15,000 children participating from kindergarten to the sixth grade.³ The purpose of the study was to determine the effect of typewriting as a supplement to handwriting and to secure evidence to support or reject adoption of regular use of the typewriter in the elementary classroom. In informal fashion, with almost no instruction in how to type, and without the touch system, the children's output of written material increased significantly when given typewriters to use, while the students suffered no loss of

¹Ibid., p. 23.

²Ralph Haefner, The Typewriter in the Primary and Intermediate Grades (New York: The Macmillan Company, 1932), p. 18.

³Benjamin D. Wood and Frank N. Freeman, An Experimental Study of the Educational Influences of the Typewriter on the Elementary School Classroom (New York: The Macmillan Company, 1932).

handwriting skills. Evidence was found to support the following conclusions:

1. Use of the typewriter was feasible in the elementary school in the conduct of the ordinary work
2. The average typewriter speed acquired in one year was equal to the average handwriting rate in each grade, with a considerable degree of typing accuracy
3. The use of the typewriter tended to stimulate pupils to produce more written material than they otherwise would
4. Use of the typewriter did not result in any appreciable loss in handwriting quality or handwriting rate
5. The use of the typewriter probably raised the level of achievement in certain school subjects without observable loss in any subject
6. The teachers regarded the typewriter as a valuable educational instrument, and pupils looked upon it with marked favor¹

The effect on beginning reading was also noted in this classic study: Gains in reading of the pupils in the experimental group who had used the typewriter were significantly larger than those of the control group.²

The typewriter has been used specifically as an aid in teaching reading in elementary schools. According to Walters, typewriting courses can be utilized as an avenue for developing word recognition by "enforcing careful and systematic observation of words and providing beneficial repetitive drill."³

¹Ibid., p. 184.

²Ibid., pp. 48-49.

³George Lewis Walters, The Development and Refinement of Reading Skills in Business Education, Monograph 128 (Cincinnati: South-Western Publishing Co., 1975), p. 68.

As a supplement to the Wood-Freeman experiment, Unzicker set out to determine the effects of typewriting on learning to read in the first grade.¹ Like Wood and Freeman, Unzicker found that the children who used typewriters did more writing than those who relied upon pencils alone. The results of the study indicated that the beginning reading of the children who used the typewriter was slightly superior in every noticeable effect to that of the control group. Furthermore, the children in the lower ranges of intelligence were aided most by the use of typewriters; pupils in upper and middle ranges of intelligence did not benefit greatly.

Kiley improved reading comprehension, as well as vocabulary and spelling, in a ninth grade typewriting class by emphasizing independent work.²

Haehn initiated projects to develop reading skills in typewriting involving proofreading, reading directions, and adjustment of reading for passages of straight copy.³

After a study was conducted in Chicago public schools in 1972 showing a relationship between learning to typewrite and improvement of language arts skills,

¹Unzicker, The Effect of the Use of the Typewriter on Beginning Reading, p. 92.

²Margaret A. Kiley, "Improve Language Skills Through Junior High Typewriting," Business Education Forum 24 (March 1970):16-17.

³Faynelle Haehn, "Let's Have a 'Read-In' in Typewriting," Fusing Reading Skills and Content, eds. H. Alan Robinson and Ellen Lamar Thomas (Newark, Delaware: International Reading Association, 1969), pp. 69-74.

twenty-seven elementary schools in Illinois in 1974-75 used Elementary Secondary Education Act (ESEA) Title I allocations to provide "packaged" typewriting programs for students in grades five through eight whose reading achievement was 1.5 or more years below the expected achievement level based upon the number of years in school.¹

A Connecticut school taught first and second grade children to type while they learned to read in order to enable writing, spelling, and language skills to keep pace with reading skills.² The experimental group in the project gained significantly in all language arts skills, particularly in creative writing. A disadvantaged experimental group achieved significantly better performance on standardized reading tests.

A Houston school succeeded in improving reading and language arts skills in a typewriting program initiated for seventh grade students who were two or more years below grade level.³

¹Charles F. Bober, "Improving Reading Achievement Through Teaching Typewriting," Business Education Forum 29 (January 1975):16-17.

²"Pilot Project," Grade Teacher 87 (May/June 1970):41.

³Winifred O'Hara, "Typewriting and Reading?" Business Teacher 46 (November/December 1968):5.

Subsequent researchers to Wood and Freeman, such as Colahan,¹ Hutchings,² Olson-Jaskari,³ Ricketts,⁴ Erickson-Clow,⁵ Whitmill,⁶ and Ellenbogen,⁷ definitely established that touch typewriting can be taught effectively to elementary school children.

John Rowe found that the electric portable's ease of handling afforded even greater possibilities in efficient typewriting instruction at an early age.⁸ After eight weeks third and fourth grade pupils typed for one minute at rates almost four times faster than their handwriting rates.

¹Wayne John Colahan, "Touch Typewriting in Two Elementary Schools" (Master's thesis, University of Chicago, 1935).

²Mabel Maxine Hutchings, "A Study of Typewriting as Offered in Elementary Schools" (Master's project, Northwestern University at Evanston, Illinois, 1948).

³M. Adeline Olson and Sylvia Jaskari, "Grade Pupils Become Proficient Typists," Balance Sheet, October 1955, pp. 56-57.

⁴Mary Carmichael Ricketts, "Typewriting for the Lower Grades," Balance Sheet, March 1950, pp. 292-94.

⁵Lawrence W. Erickson and Cletus Clow, "The Manual Portable Typewriter as a Tool of Learning With Fifth Grade Elementary School Pupils," Balance Sheet, October 1960, pp. 52-55, 59.

⁶Lucille B. Whitmill, "Typewriting: An Integral Part of the Elementary and Middle School Curriculum," Business Education Forum (April 1973):41.

⁷Abraham Ellenbogen, "An Experiment in Teaching Typewriting to Younger Students," Business Education Forum 23 (November 1968):13-14.

⁸John L. Rowe, "Readin,' Typin,' and 'Rithmetic," Business Education World, January/February/March 1959, pp. 9-12.

Tootle sought to determine the degree to which typewriting ability could be integrated into the written communication activities of the fifth grade.¹ He found that the use of the typewriter had a statistically significant positive effect on arithmetic, handwriting speed, and handwriting quality of the students as well as a positive--but not statistically significant--effect on spelling, creative writing, and study skills in science, reading, and social studies. Teachers, school administrators, and parents of experimental students considered typewriting the strongest motivational influence with which they had ever had contact.

After experimenting with fourth grade children on the electric portable typewriter, Yuen and others corroborated the fact that language arts skills seem to be learned to a somewhat higher degree with the typewriter than without the typewriter because of the motivational factor.²

Granstaff's research, also at the fifth grade level, revealed that the benefit derived from typewriting instruction can be experienced by students of various ability levels.³

¹John C. Tootle, "Typewriting in the Written Communication Activities of the Fifth Grade," National Business Education Quarterly 31 (Fall 1962):67-68.

²Jack Yuen et al., "The Electric Portable Typewriter As an Instructional Tool in Fourth Grade Language Arts," Elementary English 39 (February 1962):106-7.

³Sarah Granstaff, "An Experimental Study to Determine the Influence of Typewriting Instruction on Language-Arts Achievement in the Elementary Grades," National Business Education Quarterly 38 (October 1969):13-14.

Artuso found that the use of the typewriter in grades four through six resulted in higher gains in language arts skills for all pupils, but girls made greater gains than did boys; experimental fifth and sixth grade boys did not surpass boys in control classes in either reading or English mechanics.¹

Babbs and Cline studied the effect of the typewriter as a means of motivating students who had displayed short attention spans, were reluctant to undertake any written assignment, had poor attendance records, and in some cases appeared to have visual perception difficulties.² The project was deemed successful in terms of creating interest; recommendations included coordination between classroom teachers and the typewriting teacher in order to enhance the usefulness of typewriting instruction.

Morrison's study, conducted in the ninth grade, compared experimental classes using typewriting materials linked to immediate interests and activities of the students with control classes using conventional typewriting materials based on vocational interests.³ In four of the six areas

¹Alfred Anthony Artuso, "Motivation of the Language Arts Program in Grades Four, Five, and Six Through Instruction in Typing" (Ed.D. dissertation, University of Southern California, 1961).

²Lynda F. Babbs and Ruth Cline, "Experimental Typing Project--Denver (Colorado) Elementary School," Journal of Business Education 46 (December 1970):103-5.

³Phyllis Caulfield Morrison, "A Study of Certain General Education Values of Typewriting in the Junior High School," National Business Education Quarterly 28 (October 1959):53.

tested (capitalization, grammar, sentence structure, and rate of composing) the percentage of experimental students making gains was greater than the percentage of control students. Typewriting skills did not suffer from the innovations in materials and emphases.

Although early research substantiated the belief that elementary school children can learn typewriting and improve language arts skills at the same time, formal programs of typewriting in the elementary school have not been implemented on a wide-scale basis. Only recently has interest revived in offering typewriting at this level, primarily because typewriting practice has been found to reinforce children's acquisition of language skills. Each year of the past several years there has been a pronounced increase in typewriting enrollment in the elementary school.¹ Bober stated, "To students, the real advantage of learning to type at the elementary school level seems to be that they can take the time to reflect and think while they compose sentences, paragraphs, and themes."² Merrick's opinion, based on research findings, is that instruction in typewriting in grades seven through nine is not justified unless

¹Rowe and Thoreson, "A Historical Development of Typewriting," p. 106.

²Gerald F. Bober, "Typewriting and Language Arts," Business Education World, January-February 1970, p. 9.

the objectives are educational rather than vocational, preceding vocational specialization by several years.¹

Vocational and Personal Typewriting

In the upper grades in the beginning of its history, the subject of typewriting was vocationally slanted. The major purpose of learning to typewrite in the 1920s was to use the typewriter as a copying machine.² Beginning in the 1930s personal-use values of the subject were recognized. As personal-use typewriting grew in popularity, textbook publishers gave increasing attention to content that stressed writing on the typewriter. In vocational classes teachers stressed accuracy and speed until the 1950s; since that time more emphasis has been placed on problem-solving and production skills.³ Today the majority of students enrolled in typewriting take the course for its personal-use values rather than vocational training.⁴ Educators recommend that all students elect typewriting--regardless of their academic, professional, or vocational aspirations.

¹Nellie L. Merrick, "Effect of Instruction in Typewriting on the Fluency and Accuracy in Written Expression," School Review 49 (January-December 1941):296.

²Rowe and Thoreson, "A Historical Development of Typewriting," p. 105.

³Ibid., p. 116.

⁴Ibid., p. 107.

Nonvocational typewriting, or so-called personal typewriting, is considered an area of general education.¹ Possibly the impetus of personal typewriting courses has prompted even more research to investigate the contributions typewriting can make to general education when the traditional course is modified to integrate related learnings such as spelling, punctuation, economic concepts, vocabulary, remedial learning, and language drills.

Spelling

Research regarding the teaching of spelling in the typewriting course has produced both positive and negligible results.

Ruetten studied tenth grade students in beginning typewriting and concluded that the treatment had a positive effect on the experimental group in improvement of their reading skills; however, there was no difference at the .05 level of significance on spelling and vocabulary scores between the control and experimental group after treatment.²

Meier reported on a ninth-grade experiment in which a remedial reading group received regular English instruction in the typewriting course. Findings indicated that

¹Nathan Krevolin, "The Contribution of Typewriting to General Education," Business Education Forum 24 (January 1970):13.

²Charles William Ruetten, "Improving Reading, Vocabulary, and Spelling Skills of Students in Tenth Grade Beginning Typewriting" (Ph.D. dissertation, Arizona State University, 1976).

spelling was successfully taught, but the integration of punctuation had little value.¹

Clark's study revealed a decided variance from Meier's results.² After a high school experimental typewriting class received instruction in spelling and punctuation, Clark concluded that there may be value obtained from integrating punctuation in the typewriting course, but little value from integrating spelling.

Kathryn Rowe found that when spelling was taught only incidentally through the use of typewriters, there seemed to be little or no value.³ However, she concluded that significant improvement in spelling could be expected with a systematic program for the teaching of spelling.

Bartholome conducted a study with junior high school typewriting students.⁴ He found that the students significantly increased spelling ability by the use of special

¹Helen M. Meier, "A Report on a Fused Typewriting-English Course" (M.B.A. course report, Chicago University, 1940) as cited by Robert B. Clark, "A Determination of the Significance of Integrating Spelling and Punctuation into the Typewriting Course" (Master's thesis, Kansas State Teachers College, Pittsburg, Kansas, 1953), pp. 8, 30.

²Robert B. Clark, "A Determination of the Significance of Integrating Spelling and Punctuation into the Typewriting Course" (Master's thesis, Kansas State Teachers College, Pittsburg, Kansas, 1953), p. 30.

³Kathryn Eldoris Rowe, "Improvement of Spelling Through Typewriting" (Master's thesis, Oregon State College, 1944).

⁴Lloyd W. Bartholome, "The Typewriter as a Tool for Improving Spelling" (Doctoral dissertation, University of California, Los Angeles, 1968).

drills designed for use with the typewriter. Furthermore, students who practiced spelling on the typewriter performed as well or better in basic typewriting skills as students who did not practice spelling with typewriters.

Hansen studied the effect of the inclusion of spelling instruction in typewriting and found that it had little or no effect.¹

Balcziak conducted research in tenth-grade typewriting classes.² Experimental typewriting groups received instruction in spelling and their mean gain scores were compared with those of control groups, which were tenth-grade English classes. While there was a significant improvement in achievement in spelling, Balcziak concluded that it could not be said that there was a significant difference in spelling ability by those students who took typewriting.

However, when Albicht investigated the effects of studying spelling through the medium of the typewriter, she concluded that the method was profitable for both first- and second-year students.³

¹Stanley W. Hansen, "An Experimental Study of the Value of Spelling Instruction Integrated With Typewriting Instruction," National Business Education Quarterly 32 (October 1963):21-22.

²Louella H. Balcziak, "The Effect of Typewriting Instruction on the Ability to Spell," National Business Education Quarterly 30 (October 1961):8.

³Sister Rita M. Albicht, "Relative Effects of Studying Spelling Kinesthetically Through the Medium of the Typewriter" (Master's thesis, University of North Dakota at Grand Forks, 1965), p. 113.

Finger studied five different aspects of the effect of teaching spelling in typewriting.¹ He proposed to compare within six high-school typewriting classes the spelling achievement of (1) first- and second-year students; (2) sophomores, juniors, and seniors; (3) boys and girls; and (4) students with and without formal spelling instruction. During the semester two classes received limited instruction covering about 500 words, two classes received intensive instruction covering about 1,300 words, and two classes received no instruction. The limited spelling instruction proved to be the most effective method in that on the final test students with intensive instruction gained an average of ten words, students with limited instruction gained an average of eleven words, and students with no instruction gained an average of three words. Second-year students averaged seven more correct words than did first-year students. Each junior averaged approximately five more correct words than did each sophomore and four more correct words than did each senior. Each girl averaged approximately five more correct words than did each boy.

Incidental Learning

According to Hamilton in his synthesis of research for the period 1949-60, findings did not support the hypothesis that nontypewriting content can be learned

¹Kendall A. Finger, "A Study of the Effectiveness of Teaching Spelling as an Integral Part of Typewriting," National Business Education Quarterly 27 (October 1958):24.

incidentally while typewriting.¹ However, in 1967 Palmer concluded that students can learn the content of materials being typed.² The conclusions were based on the findings of an experiment involving 973 students in three groups: (1) typing experimental, (2) English control, and (3) typing control. In addition to pretesting and posttesting, a post-post-test was given to measure the amount of knowledge retained.

A study conducted by Clayton revealed that students increased their understandings of economics significantly by acquiring the knowledge incidentally while typewriting timed writings containing economic concepts.³

Similar results were disclosed in a study made by Cowling.⁴ He concluded that students who are actually taught economic concepts in typewriting classes, even for a short period of time, learn more economics than do students who rely on incidental learning alone.

¹Herbert Morris Hamilton, "An Analysis and Synthesis of Research Findings in Typewriting, 1949-1960," 4 vols. (Ed.D. dissertation, Indiana University, 1976).

²Kermit G. Palmer, "Constructed Content Learning in the Typing Class," Journal of Business Education 43 (December 1967):121.

³Gerald Dean Clayton, "Incidental Learning of Economic Concepts in Beginning Typewriting Classes," National Business Education Quarterly 32 (Fall 1963):9.

⁴Richard E. Cowling, Jr., "Teaching Economics in Typewriting Classes," Business Education Forum 21 (November 1966):14.

Other researchers, such as Hanson,¹ Dawley,² and Feese,³ have incorporated economics into typewriting successfully without any loss of typewriting skill.

Baty discovered that beginning typewriting students can learn the meaning of unusual words placed in context as part of their regular typing material.⁴ Three groups were used in the investigation. One group typed from paragraphs in which the meaning of words was obvious in the context, a second group typed from paragraphs containing the same words but in which the meaning of the words was not clear, and a third group received neither of these treatments. The first group had the highest mean gain in vocabulary recognition.

Second-year typewriting students achieved a significant mean gain score in business vocabulary through both incidental and intentional methods in a study conducted by Fewes.⁵

¹Marilyn J. Hanson, "Experimental Development of Economic Understandings Through Typewriting," National Business Education Quarterly 33 (October 1964):26-27.

²Linda Tell Dawley, "Incidental and Intentional Learning of Economic Information in Beginning Typewriting" (Ph.D. dissertation, North Texas State University, 1973).

³Sharron Suzanne Michaels Feese, "Incidental Learning of Consumer Concepts in Second-Semester Typewriting" (Ph.D. dissertation, Kansas State University, 1976).

⁴Wayne Murlin Baty, "Incidental Learning of Vocabulary in Beginning Typewriting Classes," National Business Education Quarterly 28 (October 1959):7-8.

⁵Rosetta Cornelia Fewes, "An Experimental Study Analyzing Intentional and Incidental Learning of Technical Business Vocabulary in Second-Year Typewriting Classes," Business Education Forum 25 (October 1970):32.

Bane hypothesized that students in junior high school would learn incidentally twenty-five basic grammar rules by typing those rules while developing typewriting skill.¹ After testing the hypothesis, she concluded that students did not learn the grammar concepts in the process of typing them.

Fitch conducted a study in which second-year experimental typewriting students typed supplementary material built around formal grammatical usage for ten minutes each day for sixty days.² The students were told that the purpose of the materials was to help them perfect their typewriting skill; no mention of the content was made by the teacher. Anything learned by typing the material was, therefore, incidental. At the 5 percent level of confidence, the differences in gains in the experimental and control groups were not significant. His conclusion was that students cannot learn a significant amount of formal grammar incidentally in typewriting classes; the typing of an exercise does not assist a student to learn the formal grammar contained in the copy if no mention is made to the student of the nature of the content of the copy.

¹Kathryn Lee Warner Bane, "The Role of the Typewriting Teacher in Reinforcing English Skills and Knowledge of Students" (Master's thesis, San Diego State College, 1966), p. 104.

²Stanley K. Fitch, "Learning Grammar in Advanced Typewriting," Journal of Business Education 38 (November 1962):61-62.

Conversely, a study by Byford similar to that of Fitch produced positive results.¹ Experimental students using special language drills scored significantly higher on posttests and follow-up tests than did control students, and a significant amount of language drill knowledge was retained after a period of six weeks had elapsed. The purposive drills consisted of a concept, an illustrative concept sample, and student application of the concept.

English as a Foreign Language

McLeod concluded that instruction in typewriting can be an effective aid in the teaching of English as a foreign language.² A typewriting experimental group of foreign students typed language patterns which were being practiced in longhand by a control class at the same time. The experimental group developed a usable typewriting skill in thirty classroom hours and made significantly better scores than did the control group on postcourse English language tests.

Donin reported in 1975 that typewriting has been used advantageously in New York City for students to learn English as a second language.³

¹David Lent Byford, "An Investigation of Language-Drill Effectiveness in Second-Semester Typewriting at the Secondary Level," Business Education Forum 27 (October 1972):35.

²Doris G. McLeod, "An Intensive Study of the Use of Typewriting Instruction as an Aid in the Learning of English as a Foreign Language," Business Education Forum 20 (December 1965):20, 25.

³Donald M. Donin, "Innovative Typewriting in New York City Schools," Business Education Forum 29 (March 1975):12-13.

Special Education

Sister Gladis found that the educable mentally handicapped can acquire modified touch typewriting skill.¹ Her experiment utilized programmed instructional materials based on the linguistic approach. The mean scores in reading and spelling of the experimental group who typed the materials exceeded those of the control group who used pen or pencil. Statistically significant gains were also found on two motor development subtests.

Van Meter experimented with a typewriting class in junior college for students whose combined scores on the School and College Ability Test indicated a low level of academic achievement and whose language patterns showed usage of nonstandard English.² Based on her findings of statistically significant gains, she concluded that the integrated typewriting-language arts class is an effective medium for increasing language skills of disadvantaged students.

¹Sister Mary Paulette Gladis, "The Influence of Typewriting on Selected Language Arts Skills and Motor Development of the Educable Mentally Handicapped" (Ph.D. dissertation, University of North Dakota, 1970).

²Betty W. Van Meter, "An Experimental Study of Language Arts Instruction in a Junior College Typewriting Course" (Master's thesis, San Francisco State College, 1968), p. 57.

Proofreading

Wong questioned the commonly-held assumption that inferior spelling ability is a major cause of poor proofreading.¹ Her study investigated the question of whether a student could know the correct spelling of a word and fail to detect an error in the same word during proofreading. The findings revealed that students proofread at an accuracy level of 93.5 percent, while their ability to spell the incorrectly typed words was 98.98 percent. Correlation coefficients attested to the fact that although a reader may know how to spell a word, he or she may overlook an error in that same word during proofreading. The study pointed up the complexity of proofreading competency and emphasized the fact that proofreading must be taught in order that students can achieve the goal of accurate production.

Dehaven studied the effect of typewriting on seventh-grade students' ability to proofread.² Major findings of the study included: (1) In proofreading of both typewritten and hand-written compositions, a significant relationship was found between ability to recognize writing errors and intelligence, reading achievement, and quality of handwriting; and (2) students who had practice in typewriting

¹Shirley M. Wong, "Proofreading Pitfalls," Business Education Forum 29 (May 1975):16-17.

²Edna Ruth Pike Dehaven, "The Effect of Typewriting on Seventh Grade Students' Ability to Recognize Composition Errors" (Doctoral dissertation, University of Oregon, 1969).

their language arts assignment made a significantly greater gain in writing skills.

Simon developed proofreading materials and analyzed the effectiveness of using them in a college-level typewriting class.¹ Improvement was noted in the experimental group which used the proofreading exercises, but there was no significant difference in the scores of experimental and control groups at the .01 level.

Summary

The review of selected literature revealed that research conducted at the elementary and junior high school levels has shown that the typewriter has been used successfully to increase skills in reading, reading comprehension, vocabulary, spelling, arithmetic, handwriting, studying, creative writing, capitalization, grammar, sentence structure, rate of composing, and the quantity and quality of writing. Many studies which integrated language arts and typewriting involved considerable composition at the typewriter. In these studies students made greater gains in overall language skills than did comparable students without typewriting instruction.

The use of typewriters in preparing language arts assignments generally has been considered successful at all

¹Judith C. Simon, "The Effect of Directed Proofreading Practice on the Development of Proofreading Skill in a College-Level Typewriting Class," Journal of Business Education 52 (February 1977):240.

school levels. However, some studies pertaining to the integration of corollary learnings in the typewriting course have failed to find significant effectiveness. Research findings regarding the teaching of spelling in the typewriting course have been inconclusive; in this review five of ten investigators found negligible results.

The review revealed that the results of studies involving incidental learning of corollary material were also mixed; three of eight studies reviewed showed insignificant effectiveness. As would be expected, intentional learning was found to be superior to incidental learning.

In the review no study was found which indicated loss of typewriting skill when part of the class was devoted to learning integrated material, either incidentally or intentionally. A number of studies resulted in significant improvement in typewriting achievement in experimental groups which spent part of the class time on corollary learning.

The review pointed out that typewriting has been used successfully as a means of increasing specific skills as well as a medium for increasing skills among specific groups such as disadvantaged students, foreign students, and remedial classes.

Thus, although the evidence has not been conclusive, the use of the typewriter as an aid to learning generally has been found beneficial when used in conjunction with language arts, especially at the elementary level, where according to Hutchings:

. . . the primary purpose of teaching typewriting in the lower grades is not to develop actual ability to type-write (not that such skill, even at an early age, would not be valuable), but rather to achieve other values that seem to result from including typing instruction.¹

At the upper school levels, however, integration of learnings in the typewriting course has not always been effective.

¹Mabel M. Hutchings, "What Children Learn Besides Typing," Business Education World, June 1951, pp. 499-500.

CHAPTER III

METHODS OF PROCEDURE

Introduction

This chapter contains the design of the study, including the overall objective, setting, subjects, special considerations, schedule, treatment methods, and testing instruments.

Overall Objective

This study was primarily concerned with determining the effect of emphasizing proofreading and selected basic English rules in a beginning typewriting class at the college level. An objective of the course was to integrate with the regular coursework whenever possible supplementary materials which would provide practice in applying proofreading and the selected basic English rules.

Setting

The study was conducted at The University of Alabama Department of Business and Office Education at University, Alabama, in the city of Tuscaloosa, Alabama, during the fall semester, 1977. Two regularly scheduled classes in beginning typewriting were studied. The classes met at eight and

nine o'clock on Monday, Wednesday, and Friday for fifty minutes each of forty-three times during fifteen consecutive weeks. The classes used the same classroom, typewriters, and textbook, Personal and Professional Typing.¹

Registration for the classes was open to all schools and colleges of the university and to all undergraduate classifications, freshman through senior. Students who had previously received formal typewriting instruction other than for college credit were allowed to register for the course along with students who had never had typewriting instruction.

The class which met at eight o'clock was designated as the control group in the experiment. The control group was taught in the traditional manner by a faculty member regularly assigned to beginning typewriting. The class which met at nine o'clock was the experimental group. The experimental class was taught by the investigator, who modified the traditional course to integrate special emphasis on proofreading, capitalization, punctuation, spelling, and word division with the normal course objectives.

Subjects

On the second day of class both groups were requested to provide the information listed in table 2.

¹S. J. Wanous and James C. Bennett, Personal and Professional Typing, 4th ed. (Cincinnati: South-Western Publishing Co., 1973).

It was noted that the experimental group had a larger percentage of students with no previous typewriting experience and a larger percentage of male students than did the control group.

TABLE 2
EXPERIMENTAL AND CONTROL GROUP CHARACTERISTICS*

	Experimental	Control
<u>Typewriting Experience</u>		
Previous Instruction	17	19
No Previous Instruction	<u>14</u>	<u>5</u>
Total	<u>31</u>	<u>24</u>
<u>Classification</u>		
Freshman	10	6
Sophomore	7	5
Junior	2	8
Senior	<u>12</u>	<u>5</u>
Total	<u>31</u>	<u>24</u>
<u>Sex</u>		
Male	11	5
Female	<u>20</u>	<u>19</u>
Total	<u>31</u>	<u>24</u>

*Only students who finished the course are reported.

A pretest instrument measuring proofreading ability to recognize errors in capitalization, punctuation, spelling, and word division was administered to both groups. Pretest scores were used to consider the equivalence of the two groups. Scores were compared by the statistical technique analysis of variance, which revealed the summation shown in table 3.

TABLE 3
A COMPARISON OF PRETEST SCORES FOR
EXPERIMENTAL AND CONTROL GROUPS

Group	Number In Group	Mean	Standard Deviation	df	<u>t</u> -Ratio
Experimental	31	97.45	14.91	53	.90*
Control	24	94.21	11.81		

*Not statistically significant, $p > .05$ ($t_{.05}$ df 60 = 2.00).

The obtained t-ratio was compared with a t table, and it was determined that there was no significant difference at the .05 level between the mean achievement scores of the two groups. It was therefore concluded that there was insufficient evidence to indicate that the two groups were significantly different before the classes began relative to proofreading ability to recognize errors in capitalization, punctuation, spelling, and word division. The difference in

samples which did occur may have been due only to chance variation or error.

Special Considerations

Supplementary material was integrated whenever possible in the coursework in conjunction with the development and improvement of typewriting skills, typewriting techniques, and correct form for arrangement of typewritten material. Emphasis was placed on thinking while typewriting and the responsibility of the typist to correct errors as material was being typed. Students understood that the learning of integrated materials was expected of them, and grades were administered for learning or failure to learn integrated materials. However, no attempt was made to delineate the time devoted to supplementary materials.

Under school regulations students were allowed to withdraw from the course without penalty until the twenty-eighth class meeting. Through attrition the experimental class decreased by eight, and the control group by nine. Only those students who were enrolled in the classes for the entire semester were considered in the study.

Students were strongly urged to attend class regularly; however, no grading penalty for absences was exercised. As an incentive measure, extra points for perfect attendance or one to four absences were added to points credited for homework assignments in the final course grade.

Three students in the experimental class took the course on a pass/fail basis; they did not take the final examination since their grade averages without the examination fulfilled requirements for a passing grade. Therefore, statistical data contained only twenty-eight scores for the experimental class for the final examination.

Students were encouraged to come to class on time by awarding extra points for promptness (under four late arrivals).

Schedule

The experimental class proceeded according to the following weekly schedule:

First week - pretest and orientation

First through fifth week - basic typewriting instruction

Fourth and fifth week - centered announcement, outline, and memorandum

Sixth week - unit exam

Sixth through fifteenth week - technique development, review of keyboard, and timed material for speed development

Seventh and eighth week - manuscripts

Ninth week - unit exam

Tenth and eleventh week - personal and business letters

Twelfth week - unit exam

Twelfth and thirteenth week - tabulations

Fourteenth week - unit exam

Thirteenth and fourteenth week - graded timed writings

Fifteenth week - review and posttest

Sixteenth week - final exam

Treatment Methods

Introduction

According to Wong:

. . . Proofreading as an instructional topic where skill is developed is not found in most textbooks; even in typewriting where its application is obvious. The same situation exists in the language arts area. Although students are urged to proofread carefully, little is available in the way of prepared instructional materials for the teacher who wishes to teach methods of proofreading.¹

Before proofreading proficiency can be developed, the student must be skillfully led into the practice of thinking while typewriting. This requirement is basic to success in production typewriting, an end result of formal instruction. Thinking at the typewriter begins when students use the typewriter for writing instead of copying. Cook contends:

There is a vast difference between the person who can use the typewriter as a substitute for a pencil and the person who can use it only as a "sight typist." The former has been taught to "think and typeWRITE." The latter has been trained to "see and stroke." The former has been taught to develop his creativity, to use his typewriter as a tool.²

¹Wong, "What Are We Doing About Proofreading," p. 122.

²Fred S. Cook, "Performance Goals at the Junior High Level," Business Education Forum 27 (November 1972):21.

Zimpfer has stated:

One of the most important skills to be taught in typewriting classes is often neglected by business teachers. That skill is teaching students the ability to compose--or write--at the machine. In the quest for higher and higher straight-copy speeds and increased production rates, the process of thinking-and-typing is sometimes pushed aside in favor of building better skills in copying activities. . . . By starting with precomposition activities during the beginning weeks of the semester, the teacher can help eliminate the difficult period that many students experience in making the transition from copying printed words to composing original thoughts and ideas.¹

Zimpfer's model for teaching composition calls for a systematic approach whereby the changeover from copying to composition will evolve naturally through the use of direct dictation accompanying the learning of the keyboard. The student response level progresses from single letters, numbers, and symbols to words, phrases, sentences, and paragraphs.

Thus, precomposition and composition activities in the typewriting class are needed to facilitate the learning of integrated skills which require thinking while typewriting. Capitalization and punctuation rules can be reinforced through practice in applying the rules. Applying the rules correctly is directly related to proofreading ability to detect errors in capitalization and punctuation. Treatment methods, therefore, involved three components--composition, English mechanics, and proofreading--for special emphasis in the typewriting course.

¹Forest Zimpfer, "A Model for Building Composition Skill," Business Education Forum 31 (November 1976):14-15.

Precomposition

Supplementary materials for the experimental treatment included short precomposition techniques (see appendix A) to help the student break away from initial reliance on the printed copy. These short activities were utilized in the opening moments of classes, beginning with the second week of instruction, in addition to dictation and composition exercises provided by the textbook. The activities were integrated with learning the keyboard and development of speed and accuracy.

Capitalization and punctuation

Beginning with page 17 of the textbook, several paragraphs from the textbook were copied without capitals and punctuation marks (see appendix A) and given to each student to keep in a class folder. The students typed one paragraph per class meeting, inserting capitals and punctuation marks as required, without referring to the textbook. The students then compared their copy with that of the textbook and retyped the paragraph, making additional corrections if needed. The rule applying to each capital and mark of punctuation was given orally to the class for their notes.

After the keyboard was learned, sentences were dictated for practice in punctuation and capitalization.

A list of capitalization and punctuation rules (see appendix A) was given to the students to copy and compose an example of each rule in a complete sentence.

A textbook page of sustained skill-building paragraphs was assigned to be typed; each punctuation mark, except periods at ends of sentences, was circled and numbered. Numbered explanations of the punctuation marks were then typed.

Copy without punctuation (see appendix A) was provided the class to be punctuated and typed in correct manuscript form.

Letters without punctuation (see appendix A) were handed out to be punctuated and typed in correct letter style.

Questions (see appendix A) were dictated for answers to be composed in complete sentences using capitalization rules.

Portions of unit tests (see appendix A) contained copy to be punctuated and typed in correct form.

Spelling and word division

In addition to spelling exercises provided in the textbook, word lists (see appendix A) were given the experimental class to arrange in columns for tables.

Lists of words (see appendix A) were provided to be divided into syllables according to word division rules.

Letters containing misspelled words (see appendix A) were provided to correct and type in proper form.

A copy of word division rules (see appendix A) was provided for study; an assignment was made to arrange the rules in outline form.

Classwork and test papers were graded for compliance with word division rules.

Proofreading

In addition to the use of proofreading exercises included in the textbook, classwork and tests were graded for typographical errors. No credit for classwork was given for letters containing typographical errors.

Testing Instruments

Pretest-posttest

Development. The pretest-posttest instrument (see appendix B) was developed to comply with the format found on spelling and punctuation sections of two standardized tests, California Achievement Test and Iowa Tests of Educational Development; comparable timings were used for sections of the test.

A list of thirty-five capitalization and punctuation rules (see appendix A) was compiled after consulting authoritative sources such as Turabian,¹ Hutchinson,² Perkins,³ Schachter,⁴ and Stewart.⁵

¹Kate L. Turabian, A Manual for Writers of Term Papers, Theses, and Dissertations, 4th ed. (Chicago: The University of Chicago Press, 1973).

²Lois Irene Hutchinson, Standard Handbook for Secretaries, 8th ed. (New York: McGraw-Hill Book Company, 1976).

³W. E. Perkins, Punctuation: A Programmed Approach (Cincinnati: South-Western Publishing Co., 1972).

⁴Norman Schachter and Alfred T. Clark, Jr., English

Items were constructed to test the ability to detect errors in the use of capitalization and punctuation on the first two sections of the test.

Items for the spelling section were selected from a list of 600 words commonly used and misspelled in various types of writing.¹

A list of word division rules (see appendix A) was compiled after consulting leading authoritative sources. Thirteen items were constructed to test the ability to recognize syllabication errors.

A passage was selected for use in part A of the proofreading section utilizing eight of the "one hundred most frequently misspelled business words."² The lines of the passage were numbered and errors incorporated to test proofreading ability to detect errors in capitalization, punctuation, and spelling. Items for part B of the proofreading section were constructed to utilize selected words from a listing of "commonly confused words."³

the Easy Way (Cincinnati: South-Western Publishing Co., 1977).

⁵Marie M. Stewart et al., Business English and Communication, 4th ed. (New York: Gregg Division, McGraw-Hill Book Company, 1972).

¹Ibid., "A Spelling '600,'" p. 518-22.

²Norman W. Elliott, Rose Palmer, and Steve Rosen, Typing Power--Spelling Power, vol. 1 (Indianapolis: Bobbs-Merrill Educational Publishing, 1975).

³Stewart, Business English and Communication, pp. 20-23.

Administration. On the second class meeting the experimental and control groups were requested to cooperate with the investigator by marking the pretest. Each class was reminded that, although the results of the test would not affect the course grade, each person should make every effort to answer the test items correctly. Each group was told that the purpose of the test was to obtain data for a research study. The specific purpose of the study was not disclosed at that time or at any time during the semester. The directions of the test were read aloud by the administrator, and each section was timed as indicated.

Scoring. The answer key (see appendix B) was attached to a piece of cardboard and punched for use in scoring the tests by hand. Totals were rechecked by a second person and keypunched on data cards.

Item analysis. The test score data was analyzed through execution of "SPEC99 . . . Multiple Choice Test Scoring, Statistical Description and Item Analysis."¹ The answer marked by each student for each item was keypunched on data cards, and execution of the program provided the percentage of correct responses for each item for an index of difficulty (see appendix C). The program provided item discrimination analysis by obtaining the percentage of

¹Harry R. Barker and Barbara M. Barker, The University of Alabama Behavioral Sciences Statistics Program Library, 2nd revised ed. (University, Alabama, 1977), p. 6.

correct answers in the upper and lower 27 percent of the respondents. The difference between the percentages for the upper and lower 27 percent was obtained for an index of discrimination (see appendix C).

The item analysis revealed that the average index of difficulty for the capitalization section was 73 percent, slightly above the desired range of 40-70 percent. On all other sections the average index of difficulty fell within the desired range.

The analysis revealed that 103 of 154, or 67 percent, of the total items on the test registered a desirable range of .2 or above for an index of discrimination; while 48 of 154, or 31 percent, of the items obtained an index of .4 or above, an excellent range.

Final examination

Selection. A final examination was selected by the teachers of the experimental and control classes for administration to both groups. The examination consisted of a letter, manuscript, and table (see appendix E) to be typed in correct form with typographical errors erased and corrected. Grading standards (see appendix E) were listed.

Administration. Each teacher administered the test according to the directions stated on the exam.

Scoring. Each set of papers was graded by the respective teacher and rechecked by a second person

according to the key. The number of typographical errors was also obtained for use in statistical analysis.

Timed writings

Selection. Two five-minute timed writings were selected by each teacher from those contained in the textbook.

Administration. During the thirteenth and fourteenth weeks, two writings were timed on two different days. The better of the writings on each day was submitted by each student for a grade. The better of the two grades for each student was used for statistical analysis.

Scoring. Timed writings were scored on the basis of crediting all words typed up to the sixth error. Each student proofread the timed writings and circled typographical errors according to the standards provided in the textbook. Gross words per minute were computed according to the textbook scale. Scores were rechecked by the respective teacher and returned to each student for verification.

Summary

The purpose of the study was to determine the effect of emphasizing proofreading and selected basic English rules in a college level beginning typewriting class by comparing two classes at The University of Alabama.

A pretest-posttest instrument was developed and administered to determine the mean change achievement scores

in the area of special experimental emphasis for the two groups for statistical comparison.

The control class proceeded along a normal beginning college typewriting course; the experimental class followed the same procedure as did the control class and in addition received special instruction emphasizing proofreading, capitalization, punctuation, spelling, and word division. Precomposition activities were used to facilitate learning of integrated materials.

Typewriting tests were administered under controlled conditions in order to use the scores for statistical analysis to determine the effect of the special emphasis in the experimental class on the development of typewriting skills.

CHAPTER IV

FINDINGS OF THE STUDY

Introduction

This chapter reports the statistical analysis of the findings of the study. The first section pertains to the selection of statistical procedure for testing the hypotheses. The second section reports the findings and interpretations of statistical analysis.

Selection of Statistical Procedure

It was hypothesized that proofreading skill in recognizing errors in capitalization, punctuation, spelling, and word division could be acquired in a beginning typewriting class at the college level concomitantly with the development of basic typewriting skills. It was assumed that an increase in achievement scores on an instrument measuring proofreading ability would result if the experimental treatment were effective. Therefore, to determine the degree of effectiveness, the change in the mean score from pretest to posttest for the experimental group would be compared with the change in the mean score for the control group. The basic study hypothesis was that the mean change

score for the experimental group would be significantly greater than the mean change score of the control group.

Furthermore, it was hypothesized that the use of part of the instruction time for integrated learnings in the experimental class would not adversely affect the development of basic typewriting skills. It was assumed that if there were no significant differences between the performances of the two groups on a test measuring production skills and on a straight-copy timed writing, there would be no reason to believe that the experimental treatment had been detrimental to development of typewriting skills in the experimental class. Therefore, to determine the effect of the experimental treatment on the development of typewriting skills in the experimental class, the mean scores achieved on two typewriting tests by the two groups would be compared. The secondary study hypotheses were that there would be no significant differences between the mean scores of the experimental and control groups on a typewriting production test and on a timed writing.

The appropriate statistical technique, analysis of variance, would be applied to the test score data through execution of "ANOVØ1 . . . T-Test (Independent Groups)"¹ in the computer statistics program library at The University of Alabama. Execution of the program would produce the means,

¹Barker and Barker, Statistics Program Library, p. 113.

standard deviations, t -ratio, and degrees of freedom in each application. The t test would then be used to determine the significance of difference between the mean change scores or mean scores of the two groups by comparing the obtained t -ratios in each case with the t -table ("Critical Values of t "¹) value for the appropriate degrees of freedom.

Findings and Interpretations

Regarding the basic study hypothesis, the following null hypothesis was employed to test for significance of difference between means: There will be no significant difference between the mean change scores from pretest to posttest on an instrument measuring proofreading ability for the experimental and control groups.

Raw pretest and posttest scores (see appendix D) for both groups were used to obtain the mean scores for the comparison shown in table 4.

TABLE 4

MEAN CHANGE SCORES FROM PRETEST TO POSTTEST
FOR THE EXPERIMENTAL AND CONTROL GROUPS

Group	Pretest Mean	Posttest Mean	Mean Change
Experimental	97.45	99.32	1.87
Control	94.21	92.96	-1.25

¹Joan Welkowitz, Robert B. Ewen, and Jacob Cohen, Introductory Statistics for the Behavioral Sciences, 2nd ed. (New York: Academic Press, Inc., 1976), p. 301.

As table 4 indicates, from pretest to posttest the mean score for the experimental group increased; the control group mean score decreased.

In each group the raw pretest score for each student was subtracted from the raw posttest score to obtain the change score. The change scores for both groups were entered on data cards for execution of the computer program, which produced the statistics shown in table 5:

TABLE 5
A COMPARISON OF MEAN CHANGE SCORES FOR THE
EXPERIMENTAL AND CONTROL GROUPS

Group	Number In Group	Mean Change	Standard Deviation	Degrees of Freedom	<u>t</u> -Ratio
Experimental	31	1.87	8.37	53	1.54*
Control	24	-1.25	6.70		

*Not statistically significant, $p > .05$ ($t_{.05}$
[60 df] = 2.00). Statistically significant, $p < .20$ ($t_{.20}$
[60 df] = 1.30).

The statistical evidence indicated that the mean change score for the experimental group was greater than that of the control group, but the difference was not large enough to be significant. The obtained t-ratio of 1.54 did not exceed the t-table value of 2.00 needed for significance. Therefore, the null hypothesis could not be rejected. The

t table showed that 1.54 was significant at the .20 level of significance.

The pretest and posttest mean scores for each group were also compared for each section of the proofreading achievement test, as shown in tables 6 and 7, by execution of "ANOVO2 . . . T-Test (Matched Pairs or Subject as Own Control)".¹

TABLE 6
A COMPARISON OF EXPERIMENTAL GROUP
MEAN SCORES BY TEST SECTIONS

Section	Pretest Mean	Posttest Mean	Mean Change	df	<u>t</u> -Ratio
Capitalization	25.19	25.61	.42	30	.72
Punctuation	25.65	25.71	.06	30	.11
Spelling	15.42	16.87	1.45	30	2.65*
Word Division	6.84	7.26	.42	30	1.05
Proofreading	24.36	23.87	- .49	30	.66

*Statistically significant, $p < .05$ ($t_{.05}$ [30 df] = 2.04), $p < .02$ ($t_{.02}$ [30 df] = 2.46):⁰⁵

As table 6 shows, the experimental group mean score increased significantly on the spelling section. Gains were also made on capitalization, punctuation, and word division; but the group mean score decreased on the proofreading section, which tested proofreading ability to recognize any

¹Barker and Barker, Statistics Program Library, p. 8.

error in capitalization, punctuation, spelling, word division, or word usage.

TABLE 7
A COMPARISON OF CONTROL GROUP MEAN
SCORES BY TEST SECTIONS

Section	Pretest Mean	Posttest Mean	Mean Change	df	<u>t</u> -Ratio
Capitalization	24.25	24.00	- .25	23	.42
Punctuation	24.92	23.38	-1.54	23	3.52*
Spelling	14.04	14.50	.46	23	.68
Word Division	7.08	7.46	.38	23	1.07
Proofreading	23.92	23.63	- .29	23	.50

*Statistically significant, $p < .05$ ($t_{.05}$ [23 df] = 2.07), $p < .01$ ($t_{.01}$ [23 df] = 2.81).

Table 7 shows that the control group mean score increased on spelling and word division sections and decreased on capitalization, punctuation, and proofreading sections, with a significant decrease on the punctuation section.

The second hypothesis to be tested was: There will be no significant difference between scores of the experimental and control groups on a final examination measuring typewriting production skills. The examination grades for the two groups were compared statistically by the analysis of variance program, and the results are summarized in table 8.

TABLE 8
A COMPARISON OF FINAL EXAM GRADES FOR THE
EXPERIMENTAL AND CONTROL GROUPS

Group	Number In Group	Mean	Standard Deviation	df	<u>t</u> -Ratio
Experimental	28	87.07	9.61	50	5.74*
Control	24	68.75	12.84		

*Statistically significant, $p < .05$ ($t_{.05}$ [df 40] = 2.02), $p < .01$ ($t_{.01}$ [df 40] = 2.70).

Table 8 shows that there was a significant difference between the means of the two groups at both the .05 and .01 levels of significance. Since the hypothesis was tested in order to determine if the experimental group performed as well as did the control group on a test of typewriting production skills, the significant difference obtained in favor of the experimental group indicated a strong possibility of a positive rather than detrimental effect on production skills of the experimental treatment emphasizing proofreading and basic English rules in the experimental class.

The final examination consisted of the production of a letter, manuscript, and table. Three points were deducted from the test grade for each typographical error found on the papers. Since proofreading for typographical errors was emphasized in both classes, the number of uncorrected errors

in each group of papers (see appendix D) was obtained for statistical comparison. Table 9 presents this analysis.

TABLE 9
A COMPARISON OF TYPOGRAPHICAL ERRORS FOUND
ON FINAL EXAMINATIONS FOR THE TWO GROUPS

Group	Number In Group	Mean	Standard Deviation	df	<u>t</u> -Ratio
Experimental	28	2.29	2.59	50	1.04*
Control	24	3.08	2.90		

*Not statistically significant, $p > .05$ ($t_{.05}$ [df 40] = 2.02).

The experimental group had a total of sixty-four errors, while the control group had seventy-four. As table 9 shows, the average number of typographical errors per student in the experimental group was lower than that of the control group; however, this difference was not large enough to be statistically significant.

The third hypothesis to be tested was: There will be no significant difference between the mean scores on timed writings for the two groups. The results of the statistical comparison are presented in table 10.

TABLE 10
A COMPARISON OF TIMED WRITING MEAN SCORES FOR
THE EXPERIMENTAL AND CONTROL GROUPS

Group	Number In Group	Mean	Standard Deviation	df	<u>t</u> -Ratio
Experimental	31	37.29	9.28	53	.10*
Control	24	37.50	6.87		

*Not statistically significant, $p > .05$ ($t_{.05}$ [df 60] = 2.00).

As table 10 indicates, there was no significant difference between the mean scores on timed writings for the two groups. Since the hypothesis was tested in order to determine if the experimental group performed as well as did the control group on timed writings, the absence of a significant difference between group means indicates that there was no evidence to support the theory that the experimental treatment adversely affected the development of typewriting skills measured by the timed writing.

A research project similar to this study was conducted by the investigator to determine the effect of emphasizing proofreading and selected basic English rules in an advanced (second-semester) typewriting class at the college level. The proofreading testing instrument which was used in this study was administered to the advanced class in order to compare precourse and postcourse achievement.

Instructional materials and methods similar to those used in the beginning experimental class were employed in the advanced class.

The advanced class pretest mean score was compared with that of the beginning experimental class. It was observed that the beginning experimental class mean score exceeded the advanced class mean score, but the difference was not statistically significant, as table 11 shows:

TABLE 11
A COMPARISON OF PRETEST SCORES FOR ADVANCED
AND BEGINNING EXPERIMENTAL CLASSES

Group	Number In Group	Mean	Standard Deviation	df	<u>t</u> -Ratio
Advanced	32	95.09	16.08	61	.60*
Beginning Experimental	31	97.45	14.91		

*Not significantly significant, $p > .05$ ($t_{.05}$ [df 60] = 2.00).

Therefore, there was no evidence to support the theory that the two groups were significantly different before the classes began relative to proofreading ability measured by the instrument.

When the pretest scores were statistically compared with the posttest scores for the advanced class, the obtained mean gain score was found to be statistically significant, as shown in table 12.

TABLE 12
A COMPARISON OF PRETEST AND POSTTEST
SCORES FOR THE ADVANCED CLASS

Score	Mean	Standard Deviation	Mean Gain	df	<u>t</u> -Ratio
Pretest	95.09	16.08			
			8.00	62	2.12*
Posttest	103.09	13.99			

*Statistically significant, $p < .05$ ($t_{.05}$ [df 60] = 2.00).

The gain scores of the advanced class and beginning experimental class were statistically compared. Table 13 indicates that there was a significant difference between the mean gain scores of the two classes.

TABLE 13
A COMPARISON OF MEAN GAIN SCORES FOR THE ADVANCED
AND BEGINNING EXPERIMENTAL CLASSES

Group	Number In Group	Mean Gain	Standard Deviation	df	<u>t</u> -Ratio
Advanced	32	8.00	9.69		
				61	2.69*
Beginning Experimental	31	1.87	8.37		

*Statistically significant, $p < .05$ ($t_{.05}$ [df 60] = 2.00), $p < .01$ ($t_{.01}$ [df 60] = 2.66).

The highest individual total posttest score in the advanced class was 134, or 87 percent of 154 possible correct answers; the highest score achieved in the beginning experimental class was 123, or 80 percent correct.

As table 14 shows, the advanced class total scores increased from pretest to posttest in each section of the test.

TABLE 14
TOTAL GAIN SCORES BY TEST SECTION
FOR THE ADVANCED CLASS

Test Section	Pretest Score	Posttest Score	Gain Score
Capitalization	779	804	25
Punctuation	771	852	81
Spelling	510	601	91
Word Division	214	240	26
Proofreading	<u>769</u>	<u>802</u>	<u>33</u>
Total	3,043	3,299	256

As table 14 indicates, the largest gain score was achieved on the spelling section, while the second largest gain score was obtained on the punctuation section.

Summary

Statistical analysis of the data collected in this study revealed that there was no significant difference at the .05 level between the mean change scores of the

experimental and control groups on an instrument measuring proofreading ability to recognize errors in capitalization, punctuation, spelling, and word division. While the mean change score achieved by the experimental group exceeded that of the control group, the difference was found to be significant only at the .20 level.

The analysis of variance statistical technique revealed that the mean gain which was achieved by the experimental group on the spelling section of the instrument was significant at the .05 and .02 levels of significance. A significant decrease resulted in the control group on the punctuation section.

Hypotheses testing to determine if there was evidence to support the theory that the experimental treatment had an adverse effect on the development of typewriting skills revealed that there was no such evidence. There was no significant difference between the scores achieved by the two groups on timed writings; there was a significant difference between the mean scores in favor of the experimental group on a final examination measuring production skills, indicating the possibility of a positive rather than a detrimental effect of the experimental treatment on the performance of production skills.

A comparison of typographical errors found on final examination papers revealed that the experimental class had fewer errors than did the control class, but the difference was not large enough to be significant.

Replication of the experimental treatment in an advanced typewriting class revealed that the advanced class achieved a significant mean gain from pretest to posttest. The greatest gains were obtained on spelling and punctuation sections of the test.

CHAPTER V

SUMMARY, CONCLUSIONS, AND RECOMMENDATIONS

Introduction

This chapter is a summary of the problem of the study, need for the study, purpose of the study, review of related literature, and findings of the study; conclusions based on the findings of the study; and recommendations relative to further research concerning integrated learnings in the typewriting course.

Summary

Statement of the problem

A major problem in education today is the nationwide prevalence of deficiencies among students in writing, spelling, and English mechanics. An interdisciplinary instructional approach is needed to aid students in achieving more effective communication skills. In the business education curriculum, typewriting is a logical course for the integration of skills related to writing since it is a valuable communication skill in itself by virtue of providing copy faster and easier to read than longhand. The problem of instructional planning and methods is to integrate related

learnings without sacrificing maximum development of typewriting skills.

Need for the study

The need for reinforcement of basic English rules among students suggested a need for the study. If it can be determined that part of typewriting class periods can be devoted beneficially to the learning of skills related to writing without adversely affecting the development of typewriting skills, such evidence could be important in developing programs and materials for improving communication skills among students.

Purpose of the study

The purpose of the study was: (1) to develop materials to modify the traditional beginning typewriting course at the college level to include special emphasis on proofreading, capitalization, punctuation, spelling, and word division without deleting or deemphasizing any aspect of the traditional course content and (2) to determine the effect of this special emphasis on the achievement of both typewriting skills and proofreading skills involving capitalization, punctuation, spelling, and word division. The objective of the experimental method of instruction was to integrate the special emphasis without sacrificing maximum typewriting skill development.

Review of related literature

The review of selected related literature revealed that, although the evidence has not been conclusive, the use of the typewriter as an aid to learning generally has been found beneficial when used in conjunction with language arts; however, integration of learnings in the typewriting course has not always been effective.

Findings of the study

On an instrument measuring proofreading ability to recognize errors in capitalization, punctuation, spelling, and word division, the experimental group mean score increased from pretest to posttest; the control group mean score decreased. However, the difference between mean change scores for the two groups was not significant at the .05 level.

When a comparison was made of the pretest-posttest mean scores for the experimental group for each section of the test, the mean gain from pretest to posttest on the spelling section was significant at the .05 level.

The mean score of the experimental group on the final examination was greater than that of the control group. The difference between the group mean scores was significant at both the .05 and the .01 levels.

There was no significant difference between mean scores on timed writings for the two groups.

Replication of the experimental treatment in an advanced typewriting class revealed that the advanced class

achieved a significant mean gain from pretest to posttest. The greatest gains were obtained on spelling and punctuation sections of the test.

Conclusions

After the statistical findings of the study were considered, it was concluded that special emphasis on proof-reading and selected basic English rules in a beginning typewriting class at the college level had a positive effect, although the improvement noted was not statistically significant at the .05 level of significance.

Since a significant gain was obtained in the experimental group on the spelling section of the pretest-posttest, the conclusion was drawn that greater effectiveness might have resulted in the experimental class if the special emphasis had been restricted to a single component such as punctuation or spelling, rather than a cluster of skills.

Since findings revealed no significant difference between the performances of the two groups on timed writings, it was concluded that there was no reason to believe that the special emphasis in the experimental group hindered development of typewriting speed and accuracy.

The significant difference found between the final examination mean scores for the two groups indicated that there was a possibility that the experimental treatment may have been a contributing factor in the achievement of a

greater mean score by the experimental group on production skills.

After significant improvement was observed in an advanced class where the experimental treatment was replicated, the conclusion was drawn that this evidence might be an indication that integrated learnings can be more easily acquired in the typewriting course after the first semester of typewriting, possibly because comparatively more time is available for supplementary instruction, and greater typewriting speed enables more practice of integrated material.

Recommendations

In view of the findings and conclusions of this study, it is recommended that the experiment be replicated in order to further test the theory presented by the study hypothesis. So important is the development of proofreading competency and the improvement of basic English rules, that research which could provide conclusive evidence concerning instructional methods would be extremely beneficial to the teaching profession.

Wong identified the necessity of teaching proofreading,¹ and Lessenberry and Robinson have stressed that improving language skills should be one of the

¹Shirley Wong, "A Study to Compare the Effects of Three Different Methods of Reading Copy When Proofreading Straight Paragraph Copy Materials by First-Year Typewriting Students" (Doctoral dissertation, Oregon State University at Corvallis, 1971).

objectives of any communication course such as typewriting.¹ According to Lacy, ". . . no matter what the typing task, proofreading is an essential ingredient. And for students to do a good job in proofreading, they must be taught the correct procedure."² Delman has stated, ". . . the ability to proofread is not innate but is learned through practice and repetition."³ Anderson believes that "altogether too many teachers do nothing to teach proofreading" and states:

The teacher should emphasize that . . . to become proficient, proofreaders must:

Spell correctly
 Know basic punctuation rules
 Know the correct application of typing usages such
 as expressing time, amounts of money, use of
 figures and symbols, etc.
 Make simple mathematical calculations
 Read for details, not merely meaning
 Be error conscious
 Know the types of errors commonly missed
 Know the places errors frequently occur⁴

According to Rowe and Thoreson, "The true test of any typing program is how well students perform on

¹D. D. Lessenberry and Jerry W. Robinson, "A Basic Vocabulary for Typewriting," Balance Sheet, May 1968, p. 388.

²Richard Lacy, "Teach Proofreading Techniques in Typewriting Courses," Business Education Forum 32 (November 1977):18.

³Douglas J. Delman, "Innovate for Built-in Relevancy in Personal Typing," Journal of Business Education 49 (October 1974):40.

⁴Ruth I. Anderson, "Proofreading: A Must in Word Processing," Journal of Business Education 52 (November 1976):60.

production assignments."¹ Production proficiency is highly dependent on proofreading competency.

Declining national test scores and other symptoms of deficiencies in writing abilities at all school levels have pointed to the need for all teachers to assume the responsibility of providing their students with reinforcement of composition skills and English mechanics whenever possible. Therefore, further research regarding the integration of learnings such as proofreading and basic English rules in the typewriting course is vitally needed.

It is also recommended that variations of the study be carried out at different school levels in order to determine the effects of similar treatments. Some ways of varying the experimental design would be:

1. Extend the treatment over a longer period of time
2. Confine the emphasis to a single component in the area of greatest need as determined by the pretest
3. Administer short tests throughout the experimental period
4. Use the pretest as a learning instrument: Reveal the results of the pretest to the group so that each student can know which test items were answered incorrectly; administer an equivalent test at the end of the treatment
5. Strengthen the intentional learning aspect of the treatment through more stringent grading
6. Reveal the purpose of the experimental treatment and the desired outcome
7. Select a textbook or supplementary textbook containing material to be emphasized

¹Rowe and Thoreson, "A Historical Development of Typewriting," p. 121.

8. Study the effects of the treatment or a similar treatment in an individualized program

The findings of the study seem to justify the recommendation that proofreading and basic English rules be recognized as integral components of advanced typewriting courses.

It is further recommended that if effectiveness is not achieved on the first attempt at integrating associated learning in the typewriting class, the experiment be repeated with successive delimitations of emphases until positive results are obtained. Research experience has clearly indicated that the integration of corollary learnings in the typewriting course has been successful only part of the time. Erickson's statement of 1963 regarding the use of the typewriter as a learning tool is still applicable today:

. . . Research evidence to date is inconclusive and contradictory; however, there is growing evidence to indicate that the typewriter can be an aid to learning in such areas as basic English and grammar, reading, spelling, and vocabulary building. . . . Much additional research . . . is needed in these areas of possible uses of the typewriter as a learning aid.¹

¹Lawrence W. Erickson, "Desired Outcomes in the Development of Other General Education Values in Business Courses," New Perspectives in Education for Business, National Business Education Association Yearbook, 1963, p. 328.

APPENDIX A
SUPPLEMENTARY MATERIALS

Precomposition

(For use after textbook page 31)

Directions: Type from dictation.

1 and 3 and 7 and 5 and 9 and 4 and 8

3 and 5 and 1 and 7 and 4 and 9 and 8

7 and 8 and 1 and 5 and 3 and 4 and 1

5 and 4 and 1 and 7 and 9 and 3 and 8

37 and 59 and 48 and 13 and 73 and 95 and 84 and 51

49 and 15 and 35 and 71 and 83 and 17 and 81 and 38

(For use after textbook page 35)

1 to 2 to 3 to 4 to 5 to 6 to 7 to 8 to 9 to 10

2 to 3 to 4 to 5 to 6 to 7 to 8 to 9 to 10 to 11

3 by 4 by 5 by 6 by 7 by 8 by 9 by 10 by 11 by 12

4 by 5 by 6 by 7 by 8 by 9 by 10 by 11 by 12 by 13

10 to 20 to 30 to 40 to 50 to 60 to 70 to 80 to 90 to 100

20 by 30 by 40 by 50 by 60 by 70 by 80 by 90 by 100 by 110

(For use after textbook page 37)

Directions: Typewrite by 2's from 2¢ to 20¢.

(2¢ 4¢ 6¢ 8¢ 10¢ 12¢ 14¢ 16¢ 18¢ 20¢)

Typewrite by 5's from 5 to 100.

(5 10 15 20 25 30 35 40 45 50 55 60 65 70 75 80 85 90 95 100)

Typewrite the odd numbers from 1 to 35.

(1 3 5 7 9 11 13 15 17 19 21 23 25 27 29 31 33 35)

Typewrite by 10's from \$10 to \$100.

(\$10 \$20 \$30 \$40 \$50 \$60 \$70 \$80 \$90 \$100)

Directions: Type the name of a food often associated with the following:

Bread and . . . (butter)	Milk and . . . (cookies)
Ice cream and . . . (cake)	Peanut butter and . . . (jelly)
Bacon and . . . (eggs)	Rice and . . . (gravy)
Sausage and . . . (biscuits)	Lettuce and . . . (tomato)
Soup and . . . (crackers)	Salt and . . . (pepper)
Ham and . . . (eggs)	Sugar and . . . (cream)

Directions: Type an opposite word to the following:

hot	off	cold	on
dark	stop	light	go
summer	love	winter	hate
work	happy	play	sad
inside	young	outside	old
fast	short	slow	tall
little	early	big	late
difficult	weak	easy	strong

Directions: Type these short sentences from dictation. (Capitalize and punctuate.)

Go to town.	Drive the car.
Cut the grass.	Write the paper.
Dance a jig.	Run a race.
Dig a hole.	Wash the car.

Ride with me.

Go to class.

Take a bath.

Bring the book.

See a movie.

Cook an egg.

Fix a flat.

Make a cake.

Typewrite the ordinals in words and figures:

(first, second, third, fourth, fifth, sixth, seventh, eighth, ninth, tenth) (1st, 2nd, 3rd, 4th, 5th, 6th, 7th, 8th, 9th, 10th)

Typewrite the two-letter state abbreviation for each state called:

Alabama (AL)

Texas (TX)

Alaska (AK)

Washington (WA)

Arizona (AR)

California (CA)

Florida (FL)

Nevada (NV)

Georgia (GA)

Ohio (OH)

New York (NY)

Mississippi (MS)

Tennessee (TN)

Colorado (CO)

Typewrite the unabbreviated state name for each two-letter abbreviation called.

Typewrite your choice of the following:

Which would you prefer?

Steak or chicken?

Ford or Chevrolet?

Cake or pie?

Coffee or tea?

Coke or 7-Up?

Milk or water?

10¢ or \$10?

Winter or summer?

Football or basketball?

Chocolate or vanilla?

Fishing or skiing?

Blue or Red?

Directions: Type the answer to these questions in one word or number:

What year is this?

What page are we on?

What date in September is today?

How old are you?

How many states are in the United States?

How many football games has Alabama won this year?

How many days are in a year?

What day of the week is today?

Directions: Type all the words you can think of that begin with the letter "w", . . . "m", . . . "p", . . . "u".

Directions: Type all the words you can think of that rhyme with the word ball, . . . cake, . . . ring, . . . hat, . . . boy, . . . book.

Directions: Typewrite an ending to these beginning sentence clauses:

When I finish school, . . .

If it rains, . . .

Although the team played well, . . .

Even though I have a car, . . .

Since I was not in class, . . .

I heard you call, and . . .

I saw that movie, and . . .

He signed up for the course, but . . .

He ran a good race, but . . .

Directions: Compose a paragraph giving your interpretation of one of these familiar quotations:

"A rolling stone gathers no moss."

"Everyone thinks his sack heaviest."

"Blood is thicker than water."

"Necessity is the mother of invention."

Directions: Compose an application letter in rough draft and retype it in mailable form.

Capitalization and Punctuation

Directions: Type each exercise without referring to the textbook, punctuating and capitalizing as you type. Open your textbook to the page indicated, check your capitalization and punctuation, and retype the exercise.

bob will go to the zoo it may be near your store if he stops there to see you please have him call me i wish to talk to him

in the long run it is the way you work that counts if you strike the keys with short sharp strokes you will learn to type well (page 17)

1. Use a comma after a dependent clause that precedes its principal clause.

one of the points to keep in mind is that we must think well to write well a clear thought is the prime need

we must know too how to use the right word in the right spot words are tools used by those who write learn to use them

we must all learn to write well this skill is so high on the list of the things all men need to have that we should work for it with zeal (page 21)

2. Use commas to set off parenthetic words, phrases, and clauses.
3. Use a semicolon between clauses of a compound sentence when no conjunction is used.

as a rule we can do much more than we think we can do most of the time though we all do much less than we think we do

the key to a job well done is to be found in the way we start we must first set a goal then go to work we win when the job is done

all of us know at least one man who just sits and waits for his ship to come in this man waits in vain for ships come to a port in which the men work (page 23)

4. Use a comma to separate coordinate clauses joined by one of the pure conjunctions (and, but, for, or, neither, nor).

do not let a pause slow you down if you want to type fast in great part your speed will be based on the way you can make time count

size up the way you strike all the keys your strokes should be quick crisp and sharp just keep in mind that the time you lose is a tax on speed

place the book on the desk so that you can see it with ease keep your eyes on the book each false move you make has its price and that price is a low rate (page 24)

5. Words in a series are separated by commas.

you can learn to type well all right to do so you must train your mind to think as you type and your hands to do the things they should you must have a clear goal before you as you work

keep your fingers well curved and hold them just over the home keys use quick short sharp strokes hold your wrists down and steady keep your eyes on this book if you wish to type well (page 26)

6. Use a comma to point off an introductory phrase containing a verb.
7. When two or more adjectives modify a noun, separate them by commas if they bear equal relationship to the noun.

you must read to stay in the race events are occurring at such a dizzy pace that unless you read you will quickly be out of touch with the world you must do something more though than expose yourself to the printed page

look upon reading as a search for meaning one who reads well seeks that is the basic technique of reading you cannot get meaning out of words simply by looking at them any more than you can wear jewels that are locked in a case (page 28)

the experts all tell us that we should learn from the mistakes of others no one has the time to make them all himself we gain when we listen to this advice

know that failure is the only prize we will acquire in this world without hard work we must be willing to work for those things that most of us need and desire to have

a speaker talking about jobs said that if one but had the right key he could sing in any flat he implied of course that a man with a useful skill would always have a good job (page 38)

all of us make mistakes but some of us give them too much aid genius is the ability to evade work by doing a job right the very first time it has to be done

the mistakes you should avoid at this point in your typing are mistakes in techniques just be exact in the way you type you can gain good skill only if you type right

of all the faults one can have in typing those he has in stroking the keys are by far the most costly you must know that even a small flaw here will be repeated dozens of times (page 40)

can she type great she can start tomorrow please call i saw her at 10 a m jerry will leave on the 9 p m flight she brought jewelry dishes clothes and linens from spain

i checked these sizes 8 10 and 12 he left at 5 17 p m
 these traits courtesy and poise should never be forgotten
 john brought a hand carved frame for the well known painting
 (page 42)

8. Use a question mark after a direct question--not after an indirect question.
9. Use an exclamation point after a word, a phrase, or a sentence to indicate strong emotion or to carry sharp emphasis.
10. Do not space after a period within an abbreviation.
11. Use a colon to introduce a listing or a question or a long quotation.
12. Use a dash to indicate a sudden change of thought, for emphasis to set off an appositive, or to introduce the name of an author when it follows a direct quotation.

Directions: Type the sentences in exercise 6a, page 12, and exercise 6d on page 13 as they are shown in the textbook. Then retype the sentences, inserting a semicolon between sentences.

Rule: Use a semicolon between clauses of a compound sentence when no conjunction is used.

(For use after page 13)

Directions: Type these sentences from dictation, using a semicolon between clauses of a compound sentence:

Hal had a fall; he fell here.

Jack can go; he can sing.

I hit it; Jane can do that.

Directions: Type these sentences from dictation, using a comma after a dependent clause that precedes its principal clause:

If it is so, let us do that.

If I see him, I will tell him.

(For use after page 42)

Directions: Type these sentences from dictation, punctuating and capitalizing correctly.

Bring two pencils--in case one breaks.

This skill--typewriting--is valuable to everyone.

The boys--Tom and Bill--have finished school.

Please send 71 in May, 93 in June, and 57 in July.

I ordered 24 boxes, but you sent only 20.

I ordered 17 books; I received 15.

He said, "Look at this."

"Look at this," said Mary.

"Come over here," said John, "and do this."

Jack said, "Here it is."

"Bring it back," said Jane.

"I did not go," said Tom. "Did you go?"

Jane was in my class last fall.

My aunt, who lives in the North, came to see us.

If you would like to order, just fill out the enclosed card.

For your convenience in replying, a questionnaire is enclosed.

If our statement is correct, please send a check immediately.

If this meets with your approval, please sign and return the enclosed form.

Directions: Answer these questions in complete sentences.

In what section of the United States do you live?

Do you live east of the campus?

Which side of the stadium do you prefer?

In which direction will you be going from here?

What do you plan to do next summer?

Directions: Retype these sentences with correct punctuation.

When you learn the facts you will change your mind

This is one job however that you can do

Come on in the water's fine
 I finished first but I had several errors
 To qualify for the team you must know the rules
 Strike the keys with quick short strokes
 The boys were these ages eight nine and ten
 He asked if we had finished by 6 pm
 It is a well known fact
 Good luck you'll need it lots of it

Directions: Type each of the following rules and compose a complete sentence illustrating application of the rule.

Capitalize

The first word after a colon if that word begins a complete sentence.
 Remember this: Type with good techniques.
 We carry these sizes: small, medium, and large.

First, last, and all other words in titles of books, articles, periodicals, headings, and plays, except words of four letters or less used as articles, conjunctions, or prepositions.
 ("The Sound of Music," "The War Between the States")

An official title when it precedes a name.
 (President Lincoln was killed. She is the treasurer.)

Days of the week, months of the year, holidays, periods of history, and historic events. (Middle Ages, Civil War)

Geographic regions, localities, and names. (the South, Upstate New York)

A noun preceding a figure except for common nouns such as line, page, and sentence. (Turn to Chapter 12, page 2)

Type as Words

Numbers from one to ten except when used with numbers above ten, which are typed as figures.
 Was the order for four or eight books?
 Order 8 typing books and 15 English books.

A number beginning a sentence.
 Eight persons are here; 25 are at home sick.

The shorter of two numbers used together. (ten 50-gallon drums)

Isolated fractions or indefinite amounts in a sentence.

Nearly one third of the students are here.

About twenty people came to the meeting.

Names of small-numbered streets and avenues (ten and under).

1020 Sixth Street

Use an Apostrophe

Add 's to form the plural of most figures, letters, and words.

(five's, A's) To show possession: Add the apostrophe and s to

(a) a singular noun and (b) a plural noun which does not end in s.

boy's bicycle

women's shoe

Add the apostrophe and s to a proper name of one syllable which ends in s.

Jones's bill

Bess's Cafeteria

Add the apostrophe only after (a) plural nouns ending in s and

(b) a proper name of more than one syllable which ends in s or z.

Adams' home, Perez' report

Use a Colon

To introduce an enumeration or a listing.

These are my favorite poets: Shelley, Keats, and Frost.

To introduce a question or a long direct quotation.

This is the question: Did you study for the test?

Use a Comma (or Commas)

After (a) introductory words, phrases or clauses and (b) words in a series.

If you can, try to visit Chicago, St. Louis, and Dallas.

To set off short direct quotations.

She said, "If you try, you can reach your goal."

Before and after (a) words which come together and refer to the same person, thing, or idea and (b) words of direct address.

John, our class president, will give the report.

It was good to see you, Joanna, at the meeting.

To set off nonrestrictive clauses (not necessary to the meaning of the sentence), but not restrictive clauses (necessary to meaning).

Your report, which deals with the issue, is great.

The girl who just left is my sister.

To set off the year and the state when they are used in the middle of a sentence.

We were in Philadelphia, Pennsylvania, on July 4, 1966, when the accident occurred.

To separate two or more parallel adjectives (adjectives that could be separated by the word "and" instead of the comma).

The tedious, boring trip was over at last.

Use a Hyphen

To join compound adjectives before a noun which they modify as a unit.

five-year period, well-laid plans

After each word or figure in a series of words or figures that modify the same noun (suspended hyphenation).

first-, second-, and third-class reservations

To form certain compound nouns.

teacher-counselor, AFL-CIO

Use the Parentheses

To enclose parenthetical or explanatory matter and added information.

The contracts (Exhibit A) are enclosed.

To enclose identifying letters or figures in lists.

Check these factors: (1) period of time, (2) rate of pay, and (3) nature of duties.

Use a Question Mark

At the end of a sentence that is a direct question; however, use a period after a request in the form of a question, or an indirect question.

What day do you plan to leave for Mexico?

Will you please mail this letter for me.

Use Quotation Marks

To enclose direct quotations.

He said, "I'll be there at eight o'clock."

To enclose titles of articles and other parts of complete publications, short poems, song titles, television programs, and unpublished works such as theses and dissertations.

"To Tell The Truth"

Use a Semicolon

To separate two or more independent clauses in a compound sentence when the conjunction is omitted.

To err is human; to forgive, divine.

It is easy to be critical; it is not so easy to be constructive.

To separate independent clauses when they are joined by a conjunctive adverb (however, consequently, etc.)

I can go; however, I may be late.

To precede an abbreviation or word that introduces an explanatory statement.

He organized his work; for example, he put work to be done in folders of different colors to indicate degrees of urgency.

Use the Underline

With titles of complete works such as books, magazines, and newspapers. (Such titles may also be typed in ALL CAPS without the underline.)

New York Times, Harper's

Directions: Punctuate and type the following letter in block style with open punctuation. Use a 60-space line, and space 10 times after the date.

Today's Date

Dr. Harold S. Johnson
School of Engineering
The University of Alabama
University, AL 35486

Dear Dr. Johnson

You will be pleased to learn I am sure that you are attracting the largest meeting audience of any speaker our chapter of the National Association of Architects has ever sponsored. We have already received 185 reservations the number will exceed 200 we are positive.

Please let me know your flight plans for instance when does your plane arrive in Terre Haute I shall be happy to arrange your local transportation and to reserve a room for you at Holiday Inn.

I cannot tell you how gratified we are to have you visit us in Terre Haute and you may be sure Doctor Johnson that I am anticipating the pleasure of meeting you personally.

Sincerely yours

L. W. Sharp, President
Terre Haute Chapter, NAA

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Directions: Punctuate and type the following in correct form for an unbound manuscript.

Spirit Duplication

(P) Although up to 300 copies can be made from a single spirit master the spirit process is generally used for runs of 10 to 100 copies. It can be used to reproduce copies of memorandums minutes of meetings notices or other material that is to be distributed to a number of people.¹

(P) The spirit master set consists of two basic parts a master sheet and a sheet of special carbon. The use of a backing sheet will improve the consistency of the type impression. If a specially prepared master set is not available simply place the special carbon paper between the master and the backing sheet with the glossy side toward you. As you type the carbon copy will be on the back of the master sheet.

(P) For better spirit duplicator masters and copies follow these suggestions

1. Prepare a model copy of the material to be typed on the master. Check the copy for accuracy of form and typing
2. Clean the type. Inasmuch as a heavy ribbon causes broad impressions and filled-in characters the ribbon should be lightweight
3. Use a firm even stroke on a nonelectric machine. On an electric typewriter one of the lower pressure settings is usually best
4. If you make an error scrape off the letter or word on the back of the master sheet with a knife or razor blade. Rub the scraped area with a correction pencil. Then, place a small piece of carbon (torn from an unused section) under the area to be typed, with the glossy side toward you. Type over the incorrect letter or word. Remove the piece of carbon as soon as you have corrected the error

¹D. D. Lessenberry et al. Century 21 Typewriting (Cincinnati: South-Western Publishing Co., 1977), p. 427.

Directions: Type the following in correct form for a left-bound manuscript, inserting punctuation as needed.

Electronic Mail Sorting

(P) The ZIP Code system of sorting mail was begun in 1963. ZIP (Zoning Improvement Plan) divides the country into delivery units each given a five-digit number. The first digit represents one of ten geographic areas the second a certain part of a geographic area the third one of the sectional center areas for sorting mail the last two a zone number for internally zoned cities or the delivery station for smaller ones.

(P) ZIP Coding was only one step in a massive program designed to bring efficiency to the chaotic condition of mail handling. This coding system foresaw the eventual use of the Optical Character Reader (an electronic mail sorter) that depends upon a numeric language (the ZIP Code) for maximum efficiency in handling mail.

(P) The marriage of ZIP and OCR occurred in 1967 when the Optical Character Reader began full-time operation in the Detroit Post Office. Since then the OCR has been installed in about twenty major cities including New York Los Angeles and Chicago.

(P) The OCR has peculiar reading habits. For example it first scans from right to left to find the beginning of the address lines on the envelope then it reads from left to right starting with the bottom line and reading toward the top. Thus addresses on envelopes must appear within a specified "read zone."

(P) Although the OCR can read speedily and accurately state names spelled in full or abbreviated in the standard manner the U.S. Postal Service prefers the use of two-letter (without periods and spaces)

ZIP abbreviations--but only if ZIP Codes are used with them. The names standard abbreviations and special two-letter ZIP abbreviations of the ten most populous states are given below as examples:

California	Calif.	CA
New York	N. Y.	NY
Pennsylvania	Pa.	PA
Texas	Tex.	TX
Illinois	Ill.	IL
Ohio	Ohio	OH
Michigan	Mich.	MI
New Jersey	N. J.	NJ
Florida	Fla.	FL
Massachusetts	Mass.	MA

(P) The two-letter abbreviations for all states are available from the local post office in USPS Publication 59. Furthermore because businesses so often use mechanical addressing devices which limit the number of spaces that can be typed per line Publication 59 also provides standard abbreviations for city and street names.

Directions: Type the following as a left-bound manuscript, inserting punctuation as needed.

Problem 1

Employment Communication

(P) A good job is a major goal of most people. According to one source Today it is estimated that job changes are expected several times in each worker's life.¹ Employment involves three phases 1 locating a job 2 getting it and 3 keeping it.

(P) A variety of sources may be used in obtaining full or part time or summer employment. The following ones may be helpful:

1. Relatives and friends can provide job leads
2. Teachers and counselors have employment contacts
3. Local newspapers carry help wanted advertisements
4. Local employment agencies have many job listings
5. Local companies have employment offices

(P) Once a job contact is made both oral and written communication become important. As Aurner and Burtness have said Your personal skill in telling what you can do is your best insurance for job security.² Their statement implies that communication is vital in both securing and keeping a job. A personal interview is an essential step in getting a job and it requires primarily oral communication. Often an application letter is required and a data sheet is generally requested also. Each of these is a test of your skill in writing effectively.

(P) Keeping the job once you have been hired is as much a matter of personal attitude and behavior as it is of performance skill.

¹W. W. Blackledge, E. H. Blackledge, and H. J. Keily, The Job You Want--How to Get It, 2d ed. (Cincinnati: South-Western Publishing Co., 1975), p. ii.

²R. R. Aurner and P. S. Burtness, Effective English for Business Communication, 6th ed. (Cincinnati: South-Western Publishing Co., 1970), p. 501.

Directions: Type the following as the fourth and final page of a top-bound manuscript. Insert punctuation marks as needed.

Problem 2

(P) In using these various documenting techniques the writer should be very careful to see that the quoted matter is reproduced exactly as it appeared in the original source. Robinson suggests: (long quotation)

Our advice about maintaining the integrity of the quotation extends even to the preservation of grammatical errors. Fortunately there is a method available to the writer by which he can absolve himself of guilt for the presence of a misspelled word or grammatical error. The method involves the use of the Latin expression sic in brackets following the error.¹

¹David M. Robinson, Writing Reports for Management Decisions (Columbus, Ohio: Charles E. Merrill Publishing Co., 1969), p. 313.

Directions: Type the following memorandum on a full sheet in correct form, punctuating and capitalizing correctly.

To: James Smith
 From: John Jones
 Subject: Special Meeting

November 15 is our regular meeting date but on Tuesday November 24 1977 at 10 am we have scheduled a special meeting to discuss promotions etc. for the following employees managers salesmen and office assistants.

Mr. Tom Brown Jr. of Bell Brothers Inc. of Columbus Georgia will speak briefly. He is a well known authority on incentive systems. I have heard him speak before I believe our staff will benefit greatly from hearing him.

However if this time is not convenient for anyone in your department please let me know. To allow time for rescheduling please notify Miss Thomas at once.

Directions: Type the following as a left-bound manuscript, inserting punctuation as needed.

Choosing a Career (Title)

(P) One of life's greatest satisfactions is a career that "fits." In choosing your career goal therefore you should "be honest with yourself." Munschauer supports this statement when he says Of all the things that comprise a career choice the human factors and attitudes shouldn't be overlooked.¹ It is important that you choose a career you like.

(P) You cannot expect to drift into a career that fits. You must choose it deliberately. You will need to match your aptitudes interests and personal qualifications with a career that requires your abilities and interests. Feingold and Swerdloff emphasize this point in the following statement: (Long Quotation)

(P) Before you can choose anything intelligently you must have some basis for your choice. Choosing without knowing is merely taking a chance as you do when you reach into a grab bag for a prize. Given several opportunities you must know enough about each to select the

one that suits your needs better than the others stimulates your interest and tests your sense of values. Otherwise you will choose blindly.²

(P) They suggest also that choosing a career may be the first really complex adult decision you will be called upon to face. This decision will determine the pattern of your life the type of friends you have where you will live and the circumstances under which you will live. There are three prime requisites for a career decision. The first is self understanding the second is knowing the requirements of the career in which you have an interest and the third is taking the steps to achieve your career goal.³

Understanding Yourself (sub-heading)

(P) If you could see yourself as others see you the task of self understanding would not be so difficult. Tennyson suggests that it may help you clarify your self concept if you try to identify the following:

1. needs
2. interests
3. attitudes
4. values
5. work role perceptions
6. competencies⁴

Career Requirements (sub-heading)

(P) Your next step in choosing a career is to investigate the careers which seem to match your self inventory. A good place to start may be the Dictionary of Occupational Titles. When you have identified those careers which interest you and which seem to match your qualifications consult the Occupational Outlook Handbook.

Footnotes

¹John L. Munschauer, "The Anatomy of a Career," Journal of College Placement (February-March, 1970), p. 35.

²S. Norman Feingold and Sol Swerdloff, Occupations and Careers (New York: McGraw-Hill Book Co., Inc., 1969), p. 2.

³Don Dillion, "Toward Matching Personal and Job Characteristics," Occupational Outlook Quarterly (Spring, 1975), pp. 3-18.

⁴W. Wesley Tennyson, "Career Exploration," Career Education, Third Yearbook, ed. by Joel H. Magisos (Washington: American Vocational Association, 1973), p. 104.

Spelling and Word Division

Directions: Hyphenate properly any word which could be divided at the end of a line to maintain a straight right margin.

separate	accommodate
ninety	proceed
existence	dictionary
accidentally	business
occurrences	recommendation
profession	disappoint
forty	embarrass
representative	personnel
parliament	undoubtedly

Directions: Tabulate the following words in four columns, centered vertically and horizontally on the page. Leave 8 spaces between columns.

Title: WORDS FREQUENTLY MISSPELLED

Date: Current

similar	accredited	parliament	proceed
grammar	facilitate	fascinating	existence
forty	associate	accommodate	offense
fiery	profession	questionnaire	separate
receipt	innocent	recommend	ninety

Directions: Type the following table in the exact center of a full sheet of paper. Leave 10 spaces between columns.

Title: WORDS FREQUENTLY CONFUSED

Date: Current

accept	except
affect	effect
cite	site
formally	formerly
forth	fourth
its	it's
loose	lose
personal	personnel
residence	residents
their	there
your	you're

Directions: Punctuate the following letter, and correct all misspelled words. Type the letter in modified block style with indented paragraphs and mixed punctuation. Use a 50-space line; space 9 times after the date.

Mr. Ray B. Jenkins
Quality Paper, Inc.
409 Tenth Street
Birmingham, AL 35298

Dear Mr. Jenkins

(P) I am very sorry to learn that you cannot attend the Paper Production Simplification Seminar in Huntsville.

(P) Yes judging from what I have observed in similiar seminars in Montgomery and Mobile, I believe that this particular seminar focuses directly on the problems that you face in redusing expenses in your firm.

(P) Should any change of plans enable you to attend the seminar Mr. Jenkins please don't hesetate to call me. I shall be at the Broadview and I plan to arrive the Sunday before the seminar.

Sincerely yours

Martin Package Company

Manager, Systems Department

PB:

Directions: Correct all misspelled words in the following letter.
Type it in modified block style with open punctuation and
blocked paragraphs. Use a 50-space line, and space 10
times after the date.

Mr. Robert T. Jefferson
Division Manger
Wesley Manufacturing Company
1024 North 27th Avenue
Nashville, TN 42406

Dear Robert

(P) A recent buletin has been issued regarding a new system of
analysing miscelaneous errors in our seperate proccesing divisions.

(P) Affective immediately, John Barnes will be transfered to your
personel office to emplement the changes need to put the new system
into affect.

(P) Please give John your corporation and make any neccessary
arrangments to accomodate his transfer.

Yours truly

Arnold W. Moore
President

Directions: Type a sentence outline from the following rules.

WORD DIVISION

Centering typewritten material on the page is a refinement of the typewriting skill. In order to keep equal left and right margins and to keep the right margin straight, typists must sometimes divide words at ends of lines. However, if you can keep the right margin reasonably straight without dividing a word, do not divide. Dividing words correctly is the typist's responsibility.

Do not overuse word division:

1. Avoid dividing a word at the end of the first line of a page or paragraph or the last complete line of a paragraph
2. Do not divide the final word on a page or paragraph
3. Do not divide words at the ends of more than two consecutive lines

Divide words:

1. Only between syllables (interview, inter-view)
2. Only if at least three letters (and the hyphen) can be left on the first line and at least three letters can be carried to the next line (collar, col-lar)
3. Near the middle of the word if possible (introduce, intro-duce)
4. After a one-letter syllable within a word (criticism, criti-cism)
5. Only at the hyphen in a compound word already containing a hyphen (self-understanding, self-understanding)

Do not divide:

1. Words with fewer than six letters (woman)
2. Words pronounced as one syllable (strength)
3. Numbers, initials, abbreviations, or contractions (14,250; T. S. Eliot; U. S.; doesn't)

APPENDIX B
PRETEST-POSTTEST

DIAGNOSTIC EXAMINATION
WRITTEN COMMUNICATION SKILLS

Instructions

Each person will receive a test booklet and an answer sheet. Do not turn the page of the booklet until given instructions. Do not make any marks on the test booklet. Write your identification and answers on the answer sheet. Use a soft-leaded pencil and be sure to remove any erased answers thoroughly. This is a diagnostic testing instrument and will not affect your course grade; however, you should make every effort to answer the items correctly. In all sections of the test your score will be the number of items correctly marked.

On the answer sheet fill in your name, student number, classification (Freshman, Sophomore, Junior, Senior), school or college (A & S, Ed., etc.), major, and previous typewriting instruction (number of years and where received).

Turn the page of the test booklet to Section I - CAPITALIZATION.

I. CAPITALIZATION

Purpose

These items measure how well you use capital letters in written expressions.

Directions

This section contains 33 sentences. Examine each sentence for capitalization errors. If the sentence is correct as written, on the answer sheet mark the space under "Correct" (C) beside the sentence number. If the sentence contains a capital letter which should not be there or if the sentence contains an uncapitalized word which should be capitalized, on the answer sheet mark the space under "Incorrect" (I) beside the sentence number.

Example

Read the following sentences:

- A. I wrote a paper on Birmingham, Alabama.

Since all words in the sentence are capitalized correctly, beside Item A mark the space in the "Correct" column.

- B. Birmingham is the largest City in Alabama.

Since the word "City" is erroneously capitalized, beside Item B mark the space in the "Incorrect" column.

C I

A. ☒ O

B. O ☒

You have eight minutes to complete this section. If you finish the section before time is called, you may look back over items in this section (this section only), but do not proceed to the next section until you are instructed to do so.

CAPITALIZATION

1. The new highway runs east and west, by-passing the main Business District.
2. Mr. Charles A. Brown is the president of Midwest Corporation of St. Louis.
3. His Mother knew what to do: She put him to bed and called Dr. Smith, a pediatrician.
4. The President is from the south, but the Vice President is not.
5. When he was a senior in high school, John visited two of our State Institutions.
6. Our speaker, formerly of The University of Alabama, is a professor of English at George Washington University.
7. Which do you prefer to use in letters: "Yours truly" or "Very truly yours"?
8. The Governor spoke to the Legislature before it adjourned for the Summer.
9. Labor Day is a Legal Holiday--it is always the first Monday in September.
10. Sheeba, our siamese cat, demands much more attention than our dog, Fido.
11. The Atlantic charter was signed by Roosevelt and Churchill on a naval ship in the middle of the Atlantic Ocean.
12. That information can be obtained from the Chamber of Commerce or the Better Business Bureau.
13. Close your letter with either "Sincerely" or "Sincerely Yours."
14. Mark Twain once said, "Nothing so needs reforming as other people's habits."
15. A mad gunman who was roaming the streets of Birmingham was seen in a north section.
16. The case must now go to the court of appeals, and it may not be settled before next fall.
17. "If he has time," said Mother, "He will go to see the Empire State Building."

18. Decisions regarding curriculum are made by Benjamin B. Brown, our Superintendent of schools.
19. The title of Chapter X is "The War between the States."
20. Mr. Johnson has invested heavily in some Montana Copper Mines.
21. Mary introduced Professor Davis at the Honor Society meeting last Friday.
22. Mr. Cobb is a member of the senate and lives near the Washington Monument.
23. I have never traveled west of the Rockies.
24. The wife of former President Nixon was at one time a business education teacher.
25. Have you read the article entitled "The Case against War"?
26. The right of free speech is guaranteed under the constitution.
27. My question is, Can we meet the delivery date?
28. This term we will study basic principles concerning the following:
 1. capitalization
 2. punctuation
 3. spelling
29. Shorthand III and typewriting 101 are taught in the Union Building Annex.
30. "I'm glad you remembered to bring the book," said Tom. "it's due at the library."
31. with the three celebrities was a president of a big Ohio corporation.
32. Heading west from New York City, we drove over the Pulaski Skyway.
33. Her father had lived in the East for twenty years before moving here last spring.

II. PUNCTUATION

Purpose

These items measure how well you understand punctuation and how well you are able to recognize errors in the use of punctuation in written communication.

Directions

This section contains 37 sentences. Examine each sentence for punctuation errors. If the sentence is correct as written, on the answer sheet mark the space under "Correct" (C) beside the sentence number. If the sentence contains a punctuation mark which should not be there or if an incorrect punctuation mark is used or if a needed punctuation mark has been omitted, on the answer sheet mark the space under "Incorrect" (I) beside the sentence number.

Example

Read the following sentences:

- A. I wrote a paper on Birmingham, Alabama.

Since the sentence correctly contains a comma after "Birmingham" and a period after "Alabama," beside Item A mark the space in the "Correct" column.

- B. Birmingham is the largest city, in Alabama.

Since the sentence erroneously contains a comma after "city," beside Item B mark the space in the "Incorrect" column.

C I

A. ☒ O

B. O ☒

You have ten minutes to complete this section. If you finish the section before time is called, you may look back over items in this section only, but do not proceed to the next section until you are instructed to do so.

PUNCTUATION

1. You may mail us your order, or you may phone it to our home shopping service.
2. Then the problem of seniority versus ability reared its ugly head.
3. Naturally we plan to attend the meeting, but we would like advance notice of the agenda topics.
4. Mr. Mason asked whether the accident could have been avoided?
5. However, carefully you check, an error will sometimes slip by.
6. The paragraph added at the end of the report, summarized the findings.
7. All the executives--and their wives, of course,--will be invited to the dinner.
8. Please turn to Chapter IV. of your text.
9. We are shipping you the order, that you placed on June 9.
10. We telephoned our distributor in Atlanta, and asked him to visit your plant next week.
11. The seniors toured Washington and visited Mount Vernon and the Naval Academy.
12. Checking is my job operating the duplicator is yours.
13. The secretarial program includes: typing, shorthand, accounting, and adding, and calculating machines.
14. A psychologist, a businessman, and a journalist each of whom is an authority, were asked to give their opinions.
15. All routine correspondence, reports, forms, etc. may be submitted directly to Mr. White.
16. You may have an appointment Monday, or Tuesday, or Thursday.
17. The hail beat against the windows, rattled on the roof, and flattened the flowers in the garden.
18. However, difficult the problem, Mr. Gunn always seems to find a solution.
19. When you leave please, close the door.

20. He read the report, and returned it to Mr. Black.
21. To think well is admirable; to act well is more so.
22. On Friday, we shall celebrate our anniversary.
23. I should like to know whether the new type cleaner is satisfactory?
24. Would you rather spend your vacation in Miami, Florida or Honolulu, Hawaii?
25. Send the bill to Edward Moss, Sr. rather than to Edward Moss, Jr.
26. His last physical examination on March 3, 1977 showed him to be in good health.
27. The bell rang; the light flashed; the whistles blew.
28. You can expect to be happy in almost any job. If you are interested and conscientious.
29. Have you completed our monthly, income statement?
30. We are returning your affidavit dated July 1. This paper being unsuited to our needs.
31. Will you, Mr. Bell be able to attend.
32. I believe it was Bacon who wrote that "Some books are to be tasted, others swallowed, and a few chewed and digested."
33. Our director complains that his assistant is rude, and inefficient.
34. We concede that your plan has merit, nevertheless, we do not feel that we can adopt it.
35. Remember this, the student who cheats on examinations cheats only himself.
36. Both jobs can be filled by a man already working here, namely, James Jackson.
37. His positive encouraging attitude bolstered the morale of the other employees.

III. SPELLING

Purpose

These items measure how well you recognize misspelled words.

Directions

This section has 33 items containing five choices in each item. Each item includes four spelling words and a fifth choice of "None of These." In each item decide which word, if any, is spelled incorrectly and fill in the space on your answer sheet that matches the number of the misspelled word. If no word is misspelled, fill in Space 5. No item contains more than one misspelled word.

Example

Read Item A.

1	2	3	4	5
picture	rose	teacher	yellow	(None of These)

Since "picture" is the misspelled word, fill in Space 1 beside Item A.

	1	2	3	4	5
A.	☒	0	0	0	0

You have eight minutes to complete this section. If you finish the section before time is called, you may look back over items in this section (this section only), but do not proceed to the next section until you are instructed to do so.

SPELLING

	1	2	3	4	5
1.	authenticity	eminent	deceit	gorgeous	(None of These)
2.	similiar	advertise	permanent	truly	_____
3.	exaggerate	illegible	grammer	planned	_____
4.	amateur	accidently	volume	virtue	_____
5.	intercede	asociate	privilege	exhaustion	_____
6.	occurences	apparatus	sincerely	waive	_____
7.	likable	maintenance	weather	reciept	_____
8.	acreddited	necessarily	regretting	brochure	_____
9.	bankruptcy	apologise	comptroller	scandal	_____
10.	exorbitant	implement	museum	fasilitate	_____
11.	permitted	folkes	transferred	coolly	_____
12.	feiry	around	beginning	courtesy	_____
13.	inconceivable	proffesion	luxurious	yield	_____
14.	innacent	potential	interested	minimize	_____
15.	monopolize	extravagance	authorize	greivance	_____
16.	competant	gauge	mediocre	original	_____
17.	fourty	column	eliminate	grateful	_____
18.	deceive	imminent	commitment	apostrophe	_____
19.	nickel	representitive	judgment	aptitude	_____
20.	cycle	parliment	warranty	development	_____
21.	mischievous	linoleum	carburetor	facinating	_____
22.	ninth	accomodate	debtor	zealous	_____
23.	indispensible	disappoint	reversible	wholly	_____
24.	procede	statistician	eligible	dilemma	_____

	1	2	3	4	5
25.	liquefy	embarrassment	prerogative	exceed	_____
26.	relevant	cancellation	reccomended	offense	_____
27.	occasional	itemize	seperate	enterprise	_____
28.	equally	ninty	cancel	belief	_____
29.	personnel	incidently	patience	erroneous	_____
30.	freight	desirable	marvellous	sponsor	_____
31.	label	questionnaire	nuisance	vague	_____
32.	adhere	exercise	quantity	specialise	_____
33.	psychology	leisure	existance	finally	_____

IV. WORD DIVISION

Purpose

These items measure how well you recognize errors in word division.

Directions

This section has 13 items containing four choices in each item. Each item includes three divided words and a fourth choice of "None of These." In each item decide which word, if any, is divided incorrectly and fill in the space on your answer sheet that matches the number of the incorrect word. If no word is incorrect, fill in Space 4. No item contains more than one word incorrectly divided.

Example

Read Item A.

1	2	3	4
A. work-man	sophom-ore	sun-burn	(None of These)

Since "sophom-ore" is the incorrect word, fill in Space 2 beside Item A.

	1	2	3	4
A.	0	☒	0	0

You have four minutes to complete this section. If you finish the section before time is called, you may look back over items in this section (this section only), but do not proceed to the next section until you are instructed to do so.

WORD DIVISION

1	2	3	4
1. nego-tiable	undoub-tedly	nobil-ity	(None of These)
2. mod-ule	para-lyze	diff-erence	_____
3. ped-estal	necess-ary	residen-tial	_____
4. ind-ustry	requi-site	ter-minate	_____
5. tenu-ous	paci-fist	appea-rance	_____
6. conside-rable	pag-eantry	sta-dium	_____
7. public-ity	expe-riences	accor-dance	_____
8. foli-age	exis-tence	compe-tition	_____
9. calcu-lable	philoso-phy	occas-ionally	_____
10. assis-tance	calen-dar	slav-ery	_____
11. inco-herent	evapo-rate	dictio-nary	_____
12. defi-nite	stren-gth	poign-ant	_____
13. throt-tle	judi-cial	well-man-nered	_____

V. PROOFREADING

Purpose

These items measure how well you recognize errors in capitalization, punctuation, spelling, word division, and word usage in written communication.

Directions

This section contains a theme in Part A and 12 sentences in Part B. In Part A each line of the theme is a numbered item. Examine each item in Parts A and B for any error in capitalization, punctuation, spelling, word division, or word usage. If an item is correct as written, on the answer sheet mark the space under "Correct" (C) beside the item number. If an item contains an error, on the answer sheet mark the space under "Incorrect" (I) beside the item number.

Example

A. Before leaving Bill wrote a note.

Since a comma needed after "leaving" has been omitted, mark the space under "Incorrect" (I) beside Item A.

C I

A. O ☒

You have nine minutes to complete this section. If you finish the section before time is called, you may look back over items in this section only.

PROOFREADING - PART A

TYPEWRITERS*

1. Before the appearance of the typewriter, business letters were
2. written in longhand. There were no women in business then; and we
3. did not have the familiar convenience of multiple copies. People
4. believe that the first typewriter was invented in England more than
5. two hundred years ago. An American produced the first writing machine,
6. but it was not similar in appearance to the machine with which we
7. are familiar today.

8. Incidentally, president Jackson signed the patent for that first
9. American writing machine. Soon after this, inventors in separate
10. places, working independently, made improvements over the beginning
11. models. These machines were huge, and very slow to operate.

12. Christopher Sholes was the first inventor to call his machine a
13. typewriter. Incidentally, the term was later applied not to the machine
14. itself but to the operator of the machine. Sholes' machine had only
15. Capital letters and an inconvenient arrangement of the keyboard. In
16. the beginning arrangement, the keys were placed in alphabetically
17. order, and many separate changes were now made to avoid jamming. The
18. development of touch typing suggested many changes that should have
19. been made in the arrangement of the keys but unfortunately no recom-
20. mendations were adopted.

21. Mark Twain believed in the typewriter after he personally timed
22. a typist at the rate of more than fifty words for one minute. In the

*Elliott, Palmer, and Rosen, Typing Power--Spelling Power,
p. 44.

23. beginning, Twain could not beleive this achievement and asked the
24. typist to repeat the effort on three separate ocassions. Twain is
25. believed to be the first author to submit a manuscript for a book in
26. typewriten form.

PROOFREADING - PART B

27. All the officers are residence of Memphis.
28. Holly Hill is the cite chosen for the new office.
29. We need to get more stationery from the supply room.
30. He has not yet formerly accepted the appointment.
31. We are proud of the attendance record of our personal.
32. Do not loose any more sleep over that matter.
33. To be successful, a teacher must have a great deal of patients.
34. The suitcase was so tightly pact that the clothes wrinkled badly.
35. Your baggage will proceed you to Toronto.
36. Ice cream is my favorite dessert.
37. Will you give me the receipt for those cookies?
38. The first addition of the novel has been exhausted.

Name _____ Date _____

Student Number _____ School or College _____

Classification _____ Major _____

Previous Typewriting Experience _____

I. CAPITALIZATION				II. PUNCTUATION				III. SPELLING					
C	I	C	I	C	I	C	I	1	2	3	4	5	
1. O	☒	21. ☒	O	1. ☒	O	21. ☒	O	1. O	O	O	O	☒	
2. ☒	O	22. O	☒	2. ☒	O	22. O	☒	2. ☒	O	O	O	O	
3. O	☒	23. ☒	O	3. ☒	O	23. O	☒	3. O	O	☒	O	O	
4. O	☒	24. ☒	O	4. O	☒	24. O	☒	4. O	☒	O	O	O	
5. O	☒	25. O	☒	5. O	☒	25. O	☒	5. O	☒	O	O	O	
6. ☒	O	26. O	☒	6. O	☒	26. O	☒	6. ☒	O	O	O	O	
7. ☒	O	27. ☒	O	7. ☒	O	27. ☒	O	7. O	O	O	☒	O	
8. O	☒	28. O	☒	8. O	☒	28. O	☒	8. ☒	O	O	O	O	
9. O	☒	29. O	☒	9. O	☒	29. O	☒	9. O	☒	O	O	O	
10. O	☒	30. O	☒	10. O	☒	30. O	☒	10. O	O	O	☒	O	
11. O	☒	31. O	☒	11. ☒	O	31. O	☒	11. O	☒	O	O	O	
12. ☒	O	32. ☒	O	12. O	☒	32. O	☒	12. ☒	O	O	O	O	
13. O	☒	33. ☒	O	13. O	☒	33. O	☒	13. O	☒	O	O	O	
14. ☒	O			14. O	☒	34. O	☒	14. ☒	O	O	O	O	
15. ☒	O			15. O	☒	35. O	☒	15. O	O	O	☒	O	
16. O	☒			16. O	☒	36. O	☒	16. ☒	O	O	O	O	
17. O	☒			17. ☒	O	37. O	☒	17. ☒	O	O	O	O	
18. O	☒			18. O	☒			18. O	O	O	O	☒	
19. O	☒			19. O	☒			19. O	☒	O	O	O	
20. O	☒	Score ____		20. O	☒	Score ____		20. O	☒	O	O	O	

III. SPELLING

	1	2	3	4	5
21.	0	0	0	☒	0
22.	0	☒	0	0	0
23.	☒	0	0	0	0
24.	☒	0	0	0	0
25.	0	0	0	0	☒
26.	0	0	☒	0	0
27.	0	0	☒	0	0
28.	0	☒	0	0	0
29.	0	☒	0	0	0
30.	0	0	☒	0	0
31.	0	☒	0	0	0
32.	0	0	0	☒	0
33.	0	0	☒	0	0

Score ____

IV. WORD DIVISION

	1	2	3	4
1.	0	☒	0	0
2.	0	0	☒	0
3.	0	☒	0	0
4.	☒	0	0	0
5.	0	0	☒	0
6.	☒	0	0	0
7.	0	0	☒	0
8.	0	☒	0	0
9.	0	0	☒	0
10.	☒	0	0	0
11.	0	0	☒	0
12.	0	☒	0	0
13.	0	0	☒	0

Score ____

V. PROOFREADING

	C	I		C	I
1.	0	☒	21.	☒	0
2.	0	☒	22.	☒	0
3.	0	☒	23.	0	☒
4.	☒	0	24.	0	☒
5.	0	☒	25.	☒	0
6.	0	☒	26.	0	☒
7.	☒	0	27.	0	☒
8.	0	☒	28.	0	☒
9.	0	☒	29.	☒	0
10.	0	☒	30.	0	☒
11.	0	☒	31.	0	☒
12.	☒	0	32.	0	☒
13.	0	☒	33.	0	☒
14.	☒	0	34.	0	☒
15.	0	☒	35.	0	☒
16.	0	☒	36.	☒	0
17.	0	☒	37.	0	☒
18.	☒	0	38.	0	☒
19.	0	☒			
20.	☒	0			

Score ____

Total Score _____

APPENDIX C
ITEM ANALYSIS OF PRETEST-POSTTEST

PRETEST ITEM ANALYSIS
SUMMARY OF CAPITALIZATION SECTION

<u>Item</u>	<u>Index of Difficulty</u>	<u>Index of Discrimination</u>
1	.87	.00
2	.90	.05
3	.88	.19
4	.58	.33
5	.82	.29
6	.67	.24
7	.87	.28
8	.90	.05
9	.87	.09
10	.21	.28
11	.72	.23
12	.96	.05
13	.65	.43
14	.97	.05
15	.88	.05
16	.76	.62
17	.42	-.19
18	.86	.19
19	.63	.53
20	.65	.52
21	.86	.09
22	.65	.57
23	.91	.04

<u>Item</u>	<u>Index of Difficulty</u>	<u>Index of Discrimination</u>
24	.65	-.15
25	.70	.38
26	.67	.38
27	.21	.15
28	.45	.14
29	.78	.38
30	.76	.38
31	.90	.29
32	.87	.19
33	.65	.66

AVERAGE .73

18/33 (54%) = .2 or above
6/33 (18%) = .4 or above

PRETEST ITEM ANALYSIS
SUMMARY OF PUNCTUATION SECTION

<u>Item</u>	<u>Index of Difficulty</u>	<u>Index of Discrimination</u>
1	.71	-.10
2	.54	.26
3	.78	.00
4	.54	.49
5	.93	.16
6	.83	.29
7	.33	.04
8	.84	.22
9	.94	.19
10	.61	.35
11	.59	.32
12	.97	.10
13	.68	.55
14	.65	.58
15	.46	.26
16	.52	.29
17	.96	.06
18	.85	.26
19	.90	.06
20	.75	.35
21	.91	.10
22	.48	.33
23	.42	.65

<u>Item</u>	<u>Index of Difficulty</u>	<u>Index of Discrimination</u>
24	.23	.26
25	.56	.17
26	.56	.42
27	.53	.23
28	.95	.16
29	.97	.13
30	.80	.29
31	.85	.42
32	.50	.26
33	.74	.49
34	.31	.45
35	.38	.55
36	.25	.39
37	.54	.32
AVERAGE	.66	
		25/37 (67%) = .2 or above
		9/37 (24%) = .4 or above

PRETEST ITEM ANALYSIS
SUMMARY OF SPELLING SECTION

<u>Item</u>	<u>Index of Difficulty</u>	<u>Index of Discrimination</u>
1	.59	.38
2	.54	-.26
3	.54	.39
4	.09	.16
5	.97	.03
6	.36	.39
7	.59	.48
8	.88	.32
9	.66	.42
10	.08	.00
11	.64	.35
12	.78	.38
13	.80	.25
14	.93	.19
15	.45	.26
16	.31	.71
17	.55	.58
18	.57	-.09
19	.63	.58
20	.41	.55
21	.24	.22
22	.25	.42
23	.11	.23

<u>Item</u>	<u>Index of Difficulty</u>	<u>Index of Discrimination</u>
24	.22	.07
25	.04	.00
26	.55	.55
27	.45	.67
28	.26	.26
29	.11	.17
30	.38	.65
31	.18	.33
32	.70	.48
33	.17	.39
AVERAGE	.46	
		24/33 (72%) = .2 or above
		11/33 (33%) = .4 or above

PRETEST ITEM ANALYSIS
SUMMARY OF WORD DIVISION SECTION

<u>Item</u>	<u>Index of Difficulty</u>	<u>Index of Discrimination</u>
1	.68	.58
2	.59	.49
3	.68	.35
4	.82	.36
5	.75	.58
6	.78	.52
7	.06	.10
8	.12	.29
9	.52	.58
10	.13	.39
11	.61	.61
12	.68	.55
13	.12	.10
AVERAGE	.50	
		11/13 (84%) = .2 or above
		7/13 (54%) = .4 or above

PRETEST ITEM ANALYSIS
SUMMARY OF PROOFREADING SECTION

<u>Item</u>	<u>Index of Difficulty</u>	<u>Index of Discrimination</u>
1	.27	.39
2	.80	.07
3	.52	.71
4	.77	.49
5	.84	.23
6	.25	.32
7	.91	.07
8	.89	.23
9	.41	.68
10	.42	.29
11	.45	.29
12	.92	.16
13	.28	.19
14	.46	.13
15	.94	.06
16	.67	.42
17	.63	.55
18	.88	.20
19	.68	.19
20	.95	.03
21	.89	.13
22	.92	.04
23	.22	.04

<u>Item</u>	<u>Index of Difficulty</u>	<u>Index of Discrimination</u>
24	.32	.51
25	.88	.20
26	.55	.58
27	.73	.55
28	.53	.35
29	.47	.13
30	.41	.52
31	.70	.52
32	.47	.51
33	.83	.48
34	.83	.42
35	.35	.35
36	.70	.42
37	.47	.04
38	.37	.42
AVERAGE	.62	25/38 (65%) = .2 or above 15/38 (40%) = .4 or above

PRETEST ITEM ANALYSIS

TOTAL TEST ITEMS

103/154 (67%) = .2 or above

48/154 (31%) = .4 or above

APPENDIX D
STATISTICAL DATA

TOTAL TEST SCORES

EXPERIMENTAL GROUP			CONTROL GROUP		
<u>Pretest</u>	<u>Posttest</u>	<u>Change</u>	<u>Pretest</u>	<u>Posttest</u>	<u>Change</u>
95	86	- 9	110	112	2
70	86	16	92	81	-11
92	89	- 3	80	84	4
108	104	- 4	104	109	5
101	89	-12	93	97	4
96	96	0	96	94	- 2
106	104	- 2	100	104	4
98	94	- 4	76	73	- 3
120	117	- 3	103	105	2
88	97	9	86	88	2
108	115	7	88	80	- 8
79	94	15	74	82	8
104	108	4	91	88	- 3
109	99	-10	122	119	- 3
109	115	6	81	81	0
82	88	6	83	85	2
87	101	14	109	117	8
63	79	16	85	87	2
90	90	0	92	80	-12
98	94	- 4	103	95	- 8
99	97	- 2	98	88	-10
78	95	17	88	96	8
68	75	7	103	88	-15
111	112	1	104	98	- 6
110	109	- 1			
103	93	-10	<u>2,261</u>	<u>2,231</u>	<u>-30</u>
118	123	5			
94	96	2			
115	103	-12			
112	118	6			
<u>110</u>	<u>113</u>	<u>3</u>			
<u>3,021</u>	<u>3,079</u>	<u>58</u>			

INDIVIDUAL SCORES - CAPITALIZATION

EXPERIMENTAL GROUP			CONTROL GROUP		
<u>Pretest</u>	<u>Posttest</u>	<u>Change</u>	<u>Pretest</u>	<u>Posttest</u>	<u>Change</u>
26	20	-6	26	28	2
24	28	4	24	19	-5
25	26	1	20	23	3
29	25	-4	28	28	0
26	26	0	25	25	0
21	19	-2	24	25	1
25	26	1	26	26	0
26	27	1	21	20	-1
30	27	-3	25	24	-1
22	27	5	23	25	2
26	25	-1	25	20	-5
22	21	-1	25	22	-3
26	29	3	22	23	1
28	25	-3	28	30	2
25	29	4	24	22	-2
23	25	2	24	23	-1
26	24	-2	24	29	5
20	26	6	24	25	1
21	25	4	26	20	-6
29	27	-2	26	25	-1
25	27	2	26	22	-4
23	22	-1	21	26	5
18	23	5	23	23	0
27	26	-1	22	23	1
24	26	2			
30	22	-8	<u>582</u>	<u>576</u>	<u>-6</u>
26	28	2			
23	25	2			
29	28	-1			
28	30	2			
<u>28</u>	<u>30</u>	<u>2</u>			
<u>781</u>	<u>794</u>	<u>+13</u>			

INDIVIDUAL SCORES - PUNCTUATION

EXPERIMENTAL GROUP			CONTROL GROUP		
<u>Pretest</u>	<u>Posttest</u>	<u>Change</u>	<u>Pretest</u>	<u>Posttest</u>	<u>Change</u>
32	23	-9	31	31	0
12	19	7	24	20	-4
23	23	0	26	23	-3
23	23	0	22	23	1
24	20	-4	24	23	-1
25	21	-4	23	26	3
27	25	-2	28	25	-3
30	29	-1	22	20	-2
27	28	1	29	28	-1
25	19	-6	21	22	1
33	34	1	23	19	-4
22	27	5	20	22	2
28	26	-2	23	19	-4
29	27	-2	30	27	-3
28	31	3	22	23	1
25	26	1	22	20	-2
23	25	2	30	28	-2
20	21	1	23	21	-2
25	25	0	23	23	0
20	23	3	29	26	-3
27	26	-1	27	22	-5
23	26	3	26	24	-2
22	25	3	27	23	-4
31	32	1	<u>23</u>	<u>23</u>	<u>0</u>
31	29	-2			
26	27	1	<u>598</u>	<u>561</u>	<u>-37</u>
29	28	-1			
22	20	-2			
25	28	3			
27	30	3			
<u>31</u>	<u>31</u>	<u>0</u>			
<u>795</u>	<u>797</u>	<u>+2</u>			

INDIVIDUAL SCORES - SPELLING

EXPERIMENTAL GROUP			CONTROL GROUP		
<u>Pretest</u>	<u>Posttest</u>	<u>Change</u>	<u>Pretest</u>	<u>Posttest</u>	<u>Change</u>
12	15	3	19	19	0
10	12	2	17	16	- 1
21	17	- 4	10	11	1
22	24	2	23	20	- 3
17	13	- 4	13	18	5
20	23	3	14	10	- 4
18	20	2	12	17	5
14	14	0	10	10	0
24	22	- 2	15	17	2
11	14	3	12	11	- 1
18	22	4	10	13	3
13	15	2	6	13	7
20	20	0	16	12	- 4
17	18	1	20	21	1
20	20	0	11	14	3
9	11	2	10	15	5
9	19	10	17	19	2
8	14	6	13	11	- 2
11	13	2	16	11	- 5
16	16	0	16	17	1
14	16	2	10	10	0
7	14	7	10	12	2
7	9	2	18	13	- 5
16	15	- 1	19	18	- 1
19	21	2	<u>19</u>	<u>18</u>	<u>- 1</u>
13	15	2	<u>337</u>	<u>348</u>	<u>+11</u>
24	24	0			
14	16	2			
21	16	- 5			
20	19	- 1			
<u>13</u>	<u>16</u>	<u>3</u>			
<u>478</u>	<u>523</u>	<u>+45</u>			

INDIVIDUAL SCORES - WORD DIVISION

EXPERIMENTAL GROUP			CONTROL GROUP		
<u>Pretest</u>	<u>Posttest</u>	<u>Change</u>	<u>Pretest</u>	<u>Posttest</u>	<u>Change</u>
8	5	-3	7	8	1
3	8	5	5	4	-1
1	6	5	7	8	1
6	8	2	3	7	4
7	6	-1	7	7	0
8	9	1	9	9	0
7	9	2	10	10	0
6	5	-1	8	8	0
8	10	2	8	9	1
6	10	4	6	5	-1
6	6	0	5	8	3
6	6	0	4	8	4
2	6	4	6	8	2
7	5	-2	11	9	-2
9	8	-1	7	6	-1
9	9	0	5	7	2
4	6	2	10	9	-1
2	4	2	6	5	-1
8	5	-3	6	6	0
10	8	-2	7	7	0
7	9	2	10	10	0
8	7	-1	9	10	1
5	4	-1	6	3	-3
9	10	1	8	8	0
8	6	-2			
8	8	0			
10	9	-1	<u>170</u>	<u>179</u>	<u>+9</u>
10	9	-1			
10	8	-2			
6	8	2			
8	8	0			
<u>212</u>	<u>225</u>	<u>+13</u>			

INDIVIDUAL SCORES - PROOFREADING

EXPERIMENTAL GROUP			CONTROL GROUP		
<u>Pretest</u>	<u>Posttest</u>	<u>Change</u>	<u>Pretest</u>	<u>Posttest</u>	<u>Change</u>
17	23	6	27	26	-1
21	19	-2	22	22	0
22	17	-5	17	19	2
28	24	-4	28	31	3
27	24	-3	24	24	0
22	24	2	26	24	-2
29	24	-5	24	26	2
22	19	-3	15	15	0
31	30	-1	26	27	1
24	27	3	24	25	1
25	28	3	25	20	-5
16	25	9	19	17	-2
28	27	-1	24	26	2
28	24	-4	33	32	-1
27	27	0	17	16	-1
16	17	1	22	20	-2
25	27	2	28	32	4
13	14	1	19	25	6
25	22	-3	21	20	-1
23	20	-3	25	20	-5
26	19	-7	25	24	-1
17	26	9	22	24	2
16	14	-2	29	26	-3
28	29	1	32	26	-6
28	27	-1			
26	21	-5			
29	34	5	<u>574</u>	<u>567</u>	<u>-7</u>
25	26	1			
30	23	-7			
31	31	0			
30	28	-2			
<u>755</u>	<u>740</u>	<u>-15</u>			

EXPERIMENTAL GROUP TEST SECTION SCORES

	<u>Pretest</u>	<u>Posttest</u>	<u>Change</u>
Capitalization	781	794	13
Punctuation	795	797	2
Spelling	478	523	45
Word Division	212	225	13
Proofreading	<u>755</u>	<u>740</u>	<u>-15</u>
Totals	<u>3,021</u>	<u>3,079</u>	<u>58</u>

CONTROL GROUP TEST SECTION SCORES

	<u>Pretest</u>	<u>Posttest</u>	<u>Change</u>
Capitalization	582	576	- 6
Punctuation	598	561	-37
Spelling	337	348	11
Word Division	170	179	9
Proofreading	<u>574</u>	<u>567</u>	<u>- 7</u>
Totals	<u>2,261</u>	<u>2,231</u>	<u>-30</u>

TIMED WRITING SCORES

<u>Experimental Group</u>	<u>Control Group</u>
42	34
37	40
35	34
47	50
48	38
40	37
33	40
38	40
41	21
44	50
29	37
51	32
51	40
31	34
30	26
39	45
29	34
21	50
34	37
38	37
30	34
32	32
43	40
54	38
30	
56	Mean = 37.50
21	
37	
21	
32	
42	
Mean = 37.29	

FINAL EXAMINATION SCORES

<u>Experimental Group</u>	<u>Control Group</u>
93	61
96	72
64	84
97	70
90	60
76	72
92	74
72	69
90	67
87	81
97	42
88	55
88	90
83	73
96	37
59	62
86	79
79	83
90	65
88	77
90	60
97	84
86	60
90	73
95	
83	Mean = 68.75
89	
97	
Mean = 87.07	

TYPOGRAPHICAL ERRORS ON FINAL EXAMINATIONS

<u>Experimental Group</u>	<u>Control Group</u>
2	2
1	2
7	1
0	0
0	6
2	2
1	4
7	5
2	5
3	0
1	11
3	1
2	0
2	4
1	8
12	0
3	3
3	0
0	6
1	3
1	5
1	1
3	5
1	0
0	<u>0</u>
3	<u>74</u>
2	<u>74</u>
0	
<u>64</u>	

ADVANCED CLASS TOTAL TEST SCORES

<u>Pretest</u>	<u>Posttest</u>	<u>Change</u>
73	91	18
86	100	14
103	106	3
81	98	17
108	118	10
111	115	4
99	94	- 5
131	134	3
87	87	0
104	106	2
98	105	7
80	103	23
80	91	11
120	124	4
100	105	5
92	98	6
87	85	- 2
89	90	1
114	133	19
108	104	- 4
101	93	- 8
108	118	10
89	102	13
109	107	- 2
71	85	14
117	118	1
66	88	22
83	106	23
86	78	- 8
64	91	27
99	106	7
99	120	21
<u>3,043</u>	<u>3,299</u>	<u>256</u>

APPENDIX E
FINAL EXAMINATION

BEGINNING TYPEWRITING - FINAL EXAMINATION

Do not write on this examination.

Type the following problems according to the instructions preceding each problem. You have 1½ hours to complete the examination. If you finish before the time is up, please leave the room quietly. Return this examination along with your problems.

Problem 1

Type the following letter in Modified Block Style with indented paragraphs. Use Open Punctuation. Use a 60-space line. Type the date on the 13th line and leave 5 blank line spaces between the date and the inside address.

(Current date)/ Mr. and Mrs. Edward James/ 2943 Blue Haven Terrace/
Dayton, Ohio 45238/ Dear Mr. and Mrs. James/ Thank you for the
opportunity to tell you about our program of arts and crafts for adults.
(P) Leisure-time hobby classes for men and women will open at the Tri-
State Arts and Crafts Center, 2101 Madison Road, the week of September 9
and will continue through December 7, with recess Thanksgiving week.
(P) Last year, enrollment in the classes reached a total of 2,400 for
the fall, winter, and summer terms. This year we will not be able to
accommodate more than 700 adults for each session. (P) Mr. Fritz Koch is
supervisor of all hobby classes, which include drawing and painting,
portrait and figure drawing, oil painting, watercolor, toleware and tray
painting, and ceramics. (P) If you would like to know more about these
hobby classes, write or call us at the above address. Sincerely yours/
MODERN ART, INCORPORATED/ William Johnson, Director/ xx

Problem 2

Type this two-page left-bound manuscript with a long quotation, enumerated items, and a footnote as you have been instructed according to Turabian.

(Title) ADVERTISING AS A CAREER

Many people in the field of advertising are chiefly creative--the writers and artists. Some are active salesmen who sell space, time, or supplies. Others analyze and study markets, research or media to see what is going on in the world. Some manage or coordinate the work of others. Most jobs, in point of fact, call for some combination of these talents in what is considered one of the most important phases of selling.

Tonne, Simon and McGill say:

(This quotation is the beginning of a new paragraph.)

By far the most important form of sales promotion is advertising. This paid form of publicity is used to create a demand for a product and stimulate business. It is also an informative and educational force that aids the consumer in making judgments and adds to his standard of living.¹

Even though work in advertising is varied, people who have done well in this field have at least three traits in common: (Enumerate) They like to solve problems; they are all sales-minded; they are interested in people and things. Workers in other lines have these traits, too, but men and women successful in this particular field need them in large measure.

First, the person in advertising must like to solve problems. After facts have been gathered and the problem defined, a course of action must be recommended, put into effect, the results appraised and, from time to time, reappraised. Sometimes even excellent work can be improved.

Second, the person should be sales-minded. He must think in terms of persuading people, presenting ideas that will appeal to their interests.

The third outstanding trait of advertising people is an engrossing interest in mankind, as well as the workings of business. The advertising man must present selling messages in ways that draw attention, arouse interest, create a desire for the products, and persuade people to buy them. He should be a salesman with a broad background, shaped by his personal experiences, the courses he took in school, and his hobbies and other outside activities: It reflects the kind of young man he was and determines the kind of older man he will become.

(Footnote) ¹Herbert A. Tonne, Sidney I. Simon, and Esby C. McGill, Business Principles, Organization, and Management, 2nd ed. (New York: McGraw-Hill Book Company, Inc., 1963), p. 144.

Problem 3

Use a full sheet of paper. Double space the body of the tabulation. Leave 10 spaces between columns. Center the headings. Add a date line. Center vertically and horizontally. Order, as is. Turn in a pattern.

(Major Heading) Rasdorf and Tracey Manufacturing Company

(Secondary Heading) AVERAGE DAILY SALES

	<u>1963</u>	<u>1964</u>
Monday	\$6,129	\$7,610
Tuesday	5,912	7,240
Wednesday	6,307	7,433
Thursday	5,848	6,951
Friday	6,087	7,125
Saturday	6,410	7,577

Note: All problems are to be in mailable or finished form. Three points will be deducted for each typographical error; one point for each poor erasure.

GRADING KEY - FINAL EXAMINATION

	<u>Points</u>
<u>Problem 1</u> (Letter)	
Modified block style	10
Indented paragraphs	5
Open punctuation	5
Company name	5
Reference initials	<u>5</u>
	<u>30</u>
<u>Problem 2</u> (Manuscript)	
Heading (5)	
All caps	2
Centered	1
Followed by triple space	2
Margins (5)	
Top	1
Left	2
Bottom	2
Long quotation (10)	
Paragraph indention	3
Body indention	4
Single spacing	3
Enumerated items (5)	
Indented or blocked	2
Spacing	3
Footnote (10)	
Dividing line	4
Indention	3
Spacing	3
Pagination	<u>5</u>
	<u>40</u>
<u>Problem 3</u> (Table)	
Horizontal placement	10
Vertical placement	10
Major and secondary headings	5
Columnar headings	<u>5</u>
	<u>30</u>

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